MACHINE LEARNING BASED EARLY DETECTION OF CATTLE DISEASES -LUMPY SKIN DISESES

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DECLARATION

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Abstract

Lumpy Skin Disease (LSD) is a highly contagious viral disease of cattle caused by the viral Lumpy Skin Disease Virus of the genus *Capripoxvirus*. In Sri Lanka it has become a serious concern for farmers, causing reduced milk yield, weight loss and damaged hides, and it spreads quickly through biting insects. Diagnosis still relies mainly on visual inspection by veterinarians, which is subjective and, in rural areas with few veterinary officers, often comes too late to protect the rest of the herd.

This research presents the LSD detection component of CattleSense, a multimodal web-based platform for the early detection of bovine diseases. A YOLOv8s object detector first locates skin nodules in a photograph of the animal, and a ResNet50 classifier fine-tuned through transfer learning then estimates the probability of LSD for each detected region and for the image. Farmers may also complete an optional checklist of Clinical sings including enlarged lumpy nodes, elevates body temperate and reduced milk yield, which is combined with the image result through a weighted fusion (70% image, 30% symptoms). The outcome is presented as a Low, Moderate or High-risk level with practical guidance, an annotated image of the detected nodules and a downloadable PDF report for veterinary reference. The generated report is available as a downloadable PDF in both English and Sinhala.

The models were trained on cattle skin images labelled as lumpy or normal skin, using 382 images for classification and a separately annotated dataset for nodule detection. During evaluation on unseen test data, the ResNet50 classifier produced an accuracy score 93.10% and a ROC-AUC of 0.9637, while the YOLOv8s detector reached a mAP@0.5 of 11.83%, highlighting how difficult it is to localize small, irregular nodules with limited annotated data. The deployed pipeline classified all 24 detection test images correctly, although this figure is likely to be optimistic because the classifier may have seen some of them during training. Deployed as a containerized Flask microservice, the module analyses an image in under three seconds on average. These results show that combining nodule detection, image classification and symptom assessment offers a practical route to early LSD screening, particularly for farmers with limited access to veterinary services.

Keywords: Lumpy Skin Disease, Deep Learning, YOLOv8, ResNet50, Transfer Learning, Hybrid Fusion, Cattle Disease Detection, Web-based Diagnosis

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the Curve
BCE	Binary Cross-Entropy
CNN	Convolutional Neural Network
CoEAI	Centre of Excellence for Artificial Intelligence
COCO	Common Objects in Context (dataset)
CPU	Central Processing Unit
DAPH	Department of Animal Production and Health
DFL	Distribution Focal Loss
EFSA	European Food Safety Authority
ELISA	Enzyme-Linked Immunosorbent Assay
ELM	Extreme Learning Machine
FMD	Foot-and-Mouth Disease
FN	False Negative
FP	False Positive
GPU	Graphics Processing Unit
HSV	Hue, Saturation and Value
IoU	Intersection over Union
JWT	JSON Web Token
LKR	Sri Lankan Rupee
LSD	Lumpy Skin Disease
mAP	Mean Average Precision
NFR	Non-Functional Requirement
PCR	Polymerase Chain Reaction
PDF	Portable Document Format
ReLU	Rectified Linear Unit
REST	Representational State Transfer
ResNet	Residual Network
ROC	Receiver Operating Characteristic
SDG	Sustainable Development Goal
TN	True Negative
TP	True Positive
UI	User Interface
YOLO	You Only Look Once

1.INTRODUCTION

1.1 Background & Literature Survey

Lumpy Skin Disease (LSD) is an infectious viral disease affecting cattle and caused by the Lumpy Skin Disease Virus (LSDV), which belongs to the *Capripoxvirus* genus in the *Poxviridae* family [1], [2]. A most recognizable sign of the disease is the sudden appearance of firm, raised and well-defined nodules on the skin, usually between two and five centimeters across, which may cover the head, neck, limbs and much of the body. Such skin nodules typically occur alongside fever, enlarged superficial lymph nodes, eye discharge, reduced appetite and a sharp fall in milk production [1], [3]. In severe cases the lesions become necrotic, secondary bacterial infections set in, and the animal may die or remain permanently unproductive. Because the virus is carried mechanically by biting insects such as mosquitoes, stable ticks and flies, it spreads easily in warm, humid environments and is notoriously difficult to contain once it enters a region [1], [4].

For many decades LSD was confined largely to Africa. Over the past fifteen years, however, it has moved steadily into the Middle East, Eastern Europe and Asia, and it is now regarded as one of the most important transboundary diseases of cattle [1], [6]. Its arrival in South Asia, including Sri Lanka, has brought the problem directly to the doorstep of local farmers [5]. Cattle farming is an important part of the Sri Lankan rural economy, with roughly 1.2 million cattle supporting dairy production, draught power and household income, most of it on small family-run farms [5]. For these farmers, an outbreak of LSD means more than a sick animal. Milk income drops, treatment costs rise, hides lose their value, and movement and trade restrictions can follow. The Department of Animal Production and Health (DAPH) treats LSD as a notifiable disease that must be reported and contained immediately [5].

The difficulty lies less in treating the disease than in recognizing it early enough. In practice, LSD is diagnosed by a veterinarian who examines the animal and looks for the characteristic nodules and accompanying clinical signs [3]. This works well when an experienced professional is on hand, but the approach has clear weaknesses. It is influenced by the examiner's judgement and clinical experience. Early lesions, which begin as small and slightly raised patches, are easily mistaken for insect bites, dermatophilosis or allergic reactions, and are often overlooked during routine checks [3]. More importantly, veterinary officers are unevenly distributed across the country, and a

farmer in a remote area may wait days for a visit, which is precisely the period in which the disease can pass to the rest of the herd and to neighboring farms. Laboratory methods such as Polymerase Chain Reaction (PCR) and virus isolation give a definitive answer, but they are costly, slow and require equipment that is not available at farm level [4], [6]. Serological tests such as ELISA detect antibodies rather than active viruses and are of limited use in the earliest stage of infection [4].

These limitations have encouraged researchers to look at Artificial Intelligence (AI), and Deep Learning (DL) in particular, as a way of supporting earlier and more consistent screening. Convolutional Neural Networks (CNNs) are well suited to the task because LSD announces itself visually, and a photograph of an affected animal contains much of the evidence a veterinarian relies on. Genemo [10] was among the first to treat LSD as a computer vision problem, using color histograms to isolate affected skin areas and a pretrained CNN to extract deep features that were then classified with an Extreme Learning Machine, reaching an accuracy of about 90%. Saqib et al. [7] trained a MobileNetV2 network optimized with RMSProp on 793 images of healthy and diseased cattle and reported an accuracy of 95%, noting that a lightweight model of this kind could in future run on a smartphone. Saha [9] carried out a broad comparison of deep learning and classical machine learning models on 840 images, where MobileNetV2 again performed best with 96% accuracy and an AUC of 0.98; this study also used k-means clustering to segment the visibly affected area of the skin.

Senthilkumar et al. [8] extended this line of work considerably by comparing ten pretrained architectures across two datasets totaling more than 3,200 images. VGG16 and MobileNetV2 achieved accuracies of about 96%, and, importantly, the authors validated their models against images of other bovine skin conditions such as dermatophilosis, ringworm and pediculosis, which is a realistic test of whether a model has learned LSD itself rather than skin abnormality in general. Most recently, Ayon et al. [11] combined VGG16, ResNet50 and InceptionV3 in a weighted ensemble to detect LSD and Foot-and-Mouth Disease simultaneously, training on more than 10,000 expert-annotated images from farms in India, Brazil and the USA and reporting an accuracy of 98.2%.

Alongside these disease-specific studies, two general-purpose architectures are directly relevant to this research. The YOLO (You Only Look Once) family of detectors locates and labels objects in a single forward pass, which makes it fast enough for interactive use.

YOLOv8, released by Ultralights in 2023, introduced an anchor-free detection head and an improved backbone, and it has quickly become a standard choice for real-time detection tasks [12]. ResNet50, introduced by Him et al. [13], uses residual skip connections to overcome the vanishing gradient problem, allowing a 50-layer network to be trained effectively. Its ImageNet-pretrained weights transfer well to new domains, which is particularly valuable when the available dataset is small, as it usually is in veterinary imaging.

Taking together, the literature shows that deep learning can distinguish LSD-affected cattle from healthy ones with high accuracy on curated datasets. What shows far less often is how such a model would be used. Table 1.1 compares the reviewed studies with the system developed in this research across the features that matter most in a farm setting: whether the lesions are localized in the image, whether clinical signs other than the image are considered, whether the result is turned into practical guidance, and whether the work was delivered as a tool that farmers can use.

Table 1.1: Comparison of existing LSD detection studies with the proposed system

Study	Approach	Reported result	Lesion localization	Clinical symptom input	Risk guidance	Deployed user-facing tool
Genemo (2023) [10]	CNN deep features + Extreme Learning Machine	~90% accuracy	Partial (colour-based area segmentation)	No	No	No
Saqib et al. (2024) [7]	MobileNetV2 with RMSProp	95% accuracy	No	No	No	No (suggested as future work)
Saha (2024) [9]	MobileNetV2, DenseNet201, Xception and others	96% accuracy, AUC 0.98	Partial (k-means segmentation)	No	No	No
Senthilkumar et al. (2024) [8]	Comparison of ten pretrained CNNs	~96% accuracy	No	No	No	No
Ayon et al. (2026) [11]	Ensemble of VGG16, ResNet50, InceptionV3 (LSD and FMD)	98.2% accuracy	No	No	No	No
Proposed system	**YOLOv8s nodule detection + ResNet50 classification + symptom fusion**	**Reported in Chapter 3**	**Yes (bounding boxes per nodule)**	**Yes (weighted checklist)**	**Yes (Low / Moderate / High)**	**Yes (CattleSense web platform, PDF report)**

This research is conducted at the CoEAI (Centre of Excellence for Artificial Intelligence) as part of Artificial Intelligence and Data Science research cluster, in the Agriculture and Livestock industry vertically. It forms one of four disease-specific components of CattleSense, an intelligent multimodal framework designed to support the early identification of bovine diseases, alongside modules for Mastitis, Foot-and-Mouth Disease and Milk Fever. The work contributes to the United Nations Sustainable Development Goals, particularly SDG 2 (Zero Hunger) by improving the resilience od dairy farming, SDG 3 (Good Health and Well-being) through improved animal health management, and SDG 12 (Responsible Consumption and Production) by helping to minimize preventable livestock losses.

1.2 Research Gap

Although the studies reviewed above demonstrate that LSD can be recognized from images, several important gaps remain between what has been shown in research and what a farmer or veterinary officer in Sri Lanka needs.

Classification without localization. Almost every existing LSD study treats the problem as whole-image classification, producing a single label for the photograph [7], [8], [11]. The two studies that go further rely on color or clustering techniques to segment affected skin [9], [10], rather than a trained detector that identifies individual nodules. As a result, the user is told that an animal may be infected but it is not shown which lesions lead to that conclusion. A system that marks each suspected nodule is easier to trust, easier for a veterinarian to verify, and more informative about how far the disease has progressed.

Reliance on visual evidence alone. LSD is a systemic disease. Fever, swollen lymph nodes and nasal or ocular discharge often appear alongside, and sometimes before, the visible nodules [1], [3]. None of the reviewed image-based studies make use of these clinical signs, even though a farmer can observe most of them without any equipment. Ignoring them discards useful evidence, particularly when a photograph is unclear or the lesions are still small.

No decision supports the end user. Existing models report a class label or an accuracy figure, which is meaningful to a researcher but not to a farmer deciding what to do next [7], [8]. There is little work on translating a model's output into a probability that can be

understood, a level of risk, and a clear recommendation, such as whether to isolate the animal or call a veterinarian.

Research prototypes rather than usable tools. Most studies stop at reporting experimental results, and practical deployment is mentioned only as future work [7], [8]. There is no accessible web-based system that allows Sri Lankan cattle farmers to photograph an animal, receive an immediate assessment of LSD, and keep a record of the result that can be shared with a veterinary officer.

Limited relevance to the local context. The datasets behind existing models were collected in other countries, such as India, Brazil and the USA [11]. Cattle breeds, coat colors, lighting and farm environments in Sri Lanka differ, and to the best of our knowledge no LSD detection tool has been designed around the needs of Sri Lankan smallholder farmers and the country's veterinary support structure [5].

In summary, there is a need for an LSD detection system that not only classifies an image but also localizes the nodules within it, combines visual evidence with clinical signs, converts the result into clear risk-based guidance, and is delivered through a platform that farmers can use without technical knowledge.

1.3 Research Problem

Considering the gaps identified above, this study addresses the following research problems:

Lumpy Skin Disease spreads rapidly among cattle, yet its diagnosis in Sri Lanka still depends on manual visual inspection by veterinarians who are often not available when they are needed. Existing AI-based approaches classify whole images without showing where the disease is, ignore the clinical signs that farmers can readily observe, and rarely progress beyond the research stage. As a result, there is no practical means for a farmer to obtain an early, understandable and trustworthy assessment of whether an animal is likely to be infected and what should be done about it.

Figures 1.1 and 1.2 illustrate why this problem is not straightforward. Figure 1.1 shows the typical appearance of LSD, with nodules scattered over the neck, body and limbs of an affected animal. Figure 1.2 shows how varied these lesions are at closer range: small, freshly raised bumps sit alongside larger nodules and older lesions with dark, necrotic scabs, all against a coat that is itself uneven in colour and texture. This variation in size,

shape and stage makes the disease hard to judge consistently by eye, and it is also what makes automated localization of individual nodules a demanding computer vision task.



Figure 1.1: Cattle affected by Lumpy Skin Disease showing characteristic skin nodules.



Figure 1.2: close-up of LSD lesions at different stages of development

This overall problem can be broken down into four more specific questions that guided the research.

Localization challenge: How can LSD nodules, which vary widely in size, shape and colour and appear against different coat colors and backgrounds, be automatically detected and localized in an ordinary photograph taken on a farm?

Classification and probability estimation: How can the detected regions, together with the overall appearance of the skin, be converted into a reliable probability that the animal has LSD, rather than a simple yes-or-no answer?

Evidence beyond the image: How can clinical signs reported by the farmer, such as fever, swollen lymph nodes and reduced milk yield, be combined with the image-based result in a way that adds genuinely independent evidence rather than repeating what the photograph already shows?

Accessibility and decision support: How can the complete analysis be delivered to farmers with minimal technical knowledge through a web platform that presents the result as a clear risk level with practical guidance, and produces a report that can be shared with a veterinarian?

Answering these questions requires a system that brings together object detection, image classification, clinical reasoning and web-based delivery in a single pipeline. The research focuses on the design, development and evaluation of this integrated system.

1.4 Research Objectives

1.4.1 Main Objective

The primary aim of this research is to develop and implement a hybrid dep learning-based detection component for Lumpy Skin Disease in cattle as part of the CattleSense platform. The component uses a YOLOv8s model to detect and localize skin nodules in cattle images and a ResNet50 model to estimate the probability of LSD and combines this visual evidence with an optional farmer-reported clinical symptom assessment through a weighted fusion strategy. The result is presented as an LSD probability with a corresponding risk level, practical guidance, an annotated image and a downloadable report, delivered through a user-friendly web platform that supports early screening and timely veterinary intervention.

1.4.2 Specific Objectives

- **Prepare a dataset of cattle skin images for model development.** Assemble cattle skin images representing LSD-affected and healthy animals, annotate the nodules with bounding boxes for detection, and apply suitable preprocessing and

augmentation so that the models can be trained, validated and tested on separate subsets.

- **Develop a YOLOv8s-based nodule detection model.** Train a YOLOv8s object detector through transfer learning to identify and localize LSD skin nodules in cattle photographs and to provide a confidence score for each detected region.
- **Develop a ResNet50-based classification model.** Fine-tune an ImageNet-pretrained ResNet50 network to estimate the probability that an image region, or the image, shows LSD, producing a continuous probability rather than a binary label.
- **Design an integrated detection-classification pipeline.** Combine the two models so that each detected region is classified and scored and merge the region scores with a whole-image assessment to produce a single image-based LSD probability, together with an annotated image showing the detected nodules.
- **Incorporate clinical symptoms through hybrid fusion and risk stratification.** Develop a weighted, rule-based scoring of farmer-reported clinical signs that the camera cannot capture, fuse it with the image-based probability (70% image, 30% symptoms), and map the final probability to Low, Moderate and High-risk levels with appropriate guidance.
- **Access the system's performance.** Assess the detector with the use of precision, recall, F1-score and mean Average Precision (mAP), the classifier using accuracy, precision, recall, F1-score, ROC-AUC and the confusion matrix, and the complete pipeline in term of end-to-end accuracy and processing time.
- Develop and integrate a farmer-friendly web-based platform. Deploy the component as a REST API microservice within CattleSense, enabling users to upload or capture an image, complete the optional symptom checklist, view the probability, risk level, guidance and highlighted nodules, and download the assessment report as an English or Sinhala PDF for veterinary reference.

2. METHODOLOGY

2.1 Overview

The Lumpy Skin Disease (LSD) detection component was developed as a hybrid, multimodal screening system that brings together deep learning-based analysis of cattle photographs and a rule-based assessment of clinical symptoms reported by the farmer. The design follows the way the disease is recognized in the field. A veterinarian first looks at the skin for the characteristic nodules and then considers signs that cannot be seen in a photograph, such as fever or swollen lymph nodes. The methodology mirrors this reasoning in two complementary branches.

The first branch, the image analysis branch, operates in two stages. A YOLOv8s object detector searches the uploaded photograph for skin nodules and returns abounding box together with a confidence score for each identified region. Each region is then cropped and passed to a ResNet50 classifier, fine-tuned through transfer learning, which estimates the probability that the region shows LSD. The same classifier also evaluates the whole image, so that the system does not depend entirely on the detector when nodules are small, faint or spread diffusely across the skin. The region scores and the whole-image score are combined into a single image-based probability.

The second branch, the clinical symptom branch, is optional. The farmer may complete a short checklist covering swollen lymph nodes, high fever, body temperature, nasal discharge, ocular discharge, reduced milk production and decreased appetite. These signs are scored with clinically motivated weights to produce a symptom-based probability. Nodule-related questions are deliberately left out of this checklist, since the image branch already measures nodules directly and counting them twice would overstate the same evidence.

The outputs of the two branches are combined in a hybrid decision layer using a weighted fusion in which the image contributes 70% and the symptoms 30%. When no symptoms are reported, the image result is used on its own. The final probability is mapped to one of three risk levels, Low, Moderate or High, each with practical guidance for the farmer. The complete output consists of the overall LSD probability, the risk level and guidance, a breakdown of how the image and the symptoms each contributed, an annotated image in

which every detected nodule is marked, and a downloadable English or Sinhala PDF report for veterinary reference.

The overall workflow therefore runs as follows. The user uploads or captures a photograph through the CattleSense web interface and, if desired, completes the symptom checklist. The request passes through the CattleSense backend gateway to the LSD microservice, where the image is decoded, analyzed by the YOLOv8s and ResNet50 pipeline, combined with the symptom score, and returned to the user as an interpretable result that can be saved to the animal's record.

2.1.1 System Architecture

The architecture of the LSD component was designed around the principle that each stage should do one job well and pass a clearly defined output to the next. The component consists of the following modules: an image acquisition layer, a preprocessing layer, the YOLOv8s nodule detection module, the ResNet50 classification and region scoring module, the clinical symptom assessment module, the hybrid decision layer, the risk guidance module, and the output generation module responsible for the annotated image and the PDF report.

Images enter the system through the web interface, either as an uploaded file or as a photograph captured directly from the device camera. The preprocessing layer validates the file type and size, decodes the image in memory and converts it into the format expected by each model. The detection module processes the full image at 640 by 640 pixels and returns candidate nodule regions. The classification module crops each region, resizes it to 224 by 224 pixels and estimates its LSD probability, while also classifying the complete image. A region score is calculated for every detection, and the highest region score is compared with the whole-image score to obtain the image-based probability.

In parallel, the symptom assessment module converts any checklist answers into a weighted symptom probability. The hybrid decision layer fuses the two probabilities, and the risk guidance module translates the result into a risk level and recommendation. Finally, the output module draws the detected regions on the image and assembles the response. The same response can later be sent back to the service to generate a PDF report without repeating the analysis.

At deployment level, the component runs as an independent Flask microservice on port 5003 inside a Docker container. It is one of four disease microservices on the CattleSense platform. The React frontend never communicates with the microservice directly; all requests pass through the CattleSense backend gateway, which handles authentication, stores results in a PostgreSQL database and forwards analysis requests to the LSD service over a private Docker network. The deployment architecture is shown in Figure 2.8.

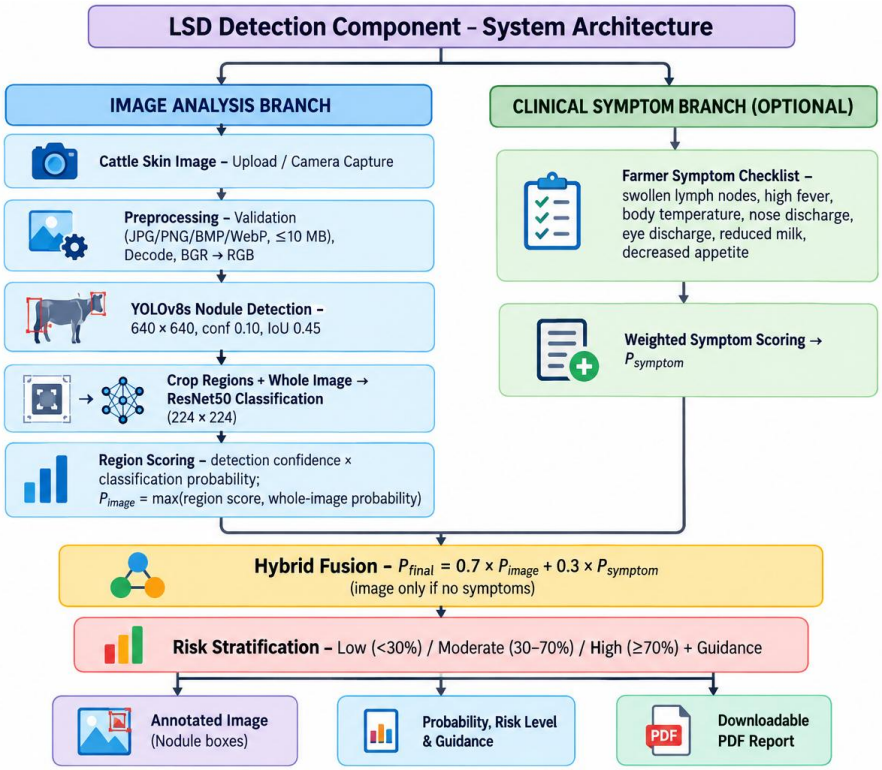


Figure 2.1: System Architecture of the LSD Detection Component

2.1.2 Data Collection and Dataset Preparation

The performance of any deep learning model is strongly influenced by the quality and organization of the data used for training. Two related but separately prepared datasets were used in this research, because the two models solve different problems. The detector needs to know the location of each nodule, which requires bounding box annotations, while the classifier only needs to determine whether an image contains LSD, which requires image-level labels.

The classification dataset was organized into two folders, Lumpy Skin and Normal Skin, containing photographs of LSD-affected and healthy cattle respectively. It comprised 382 images in total, of which 320 were LSD-positive and 62 were healthy. A stratified split was applied with a fixed random seed, allocating approximately 70% of the images to

training, 15% to validation and 15% to testing, so that both classes were represented in the same proportion in each subset. The resulting subsets contained 266 training images (222 positive, 44 healthy), 58 validation images (49 positive, 9 healthy) and 58 test images (49 positive, 9 healthy). The imbalance between the two classes reflects the material that was available and is considered when interpreting the results.

For the detection dataset, LSD nodules were annotated manually with bounding boxes using the Roboflow platform. A single class representing an LSD nodule was used, since the purpose of the detector is to locate lesions rather than to distinguish between lesion types. Annotation was carried out carefully to mark each visible nodule individually, although the task is inherently difficult: nodules vary widely in size, often merge into one another, and are sometimes only faintly visible against the coat. Images of healthy skin were retained without annotations as background images, which teach the detector what a nodule is not. The annotations were exported using the YOLO format, with each object represented by its class, the normalized coordinates of its center, and its width and height.

The detection dataset contained 254 original images, which were divided within Roboflow into 182 training, 48 validation and 24 test images. Augmentation was subsequently applied only to the training set, expanding it threefold to 546 images, including 129 background images. The validation set contained 500 annotated nodules, and the test set contained 261 annotated nodules. Six of the test images were healthy background images, which allowed the integrated pipeline to be evaluated on both positive and negative cases. Table 2.1 summarizes both datasets. Figure 2.2 shows representative samples from each class, and Figure 2.3 shows an example of the manual nodule annotation.

Table 2.1: Summary of the datasets used for detection and classification.

Attribute	Detection dataset (YOLOv8s)	Classification dataset (ResNet50)
Purpose	Localize individual LSD nodules	Estimate the probability of LSD
Label type	Bounding boxes, single class (nodule)	Image-level label (Lumpy Skin / Normal Skin)
Annotation tool	Roboflow	Folder-based labelling
Training set	182 original images, augmented to 546 (129 background)	266 images (222 LSD, 44 healthy)
Validation set	48 images, 500 nodule instances	58 images (49 LSD, 9 healthy)
Test set	24 images (18 LSD, 6 healthy), 261 nodule instances	58 images (49 LSD, 9 healthy)
Split strategy	Roboflow train / valid / test split	Stratified 70 / 15 / 15, fixed seed

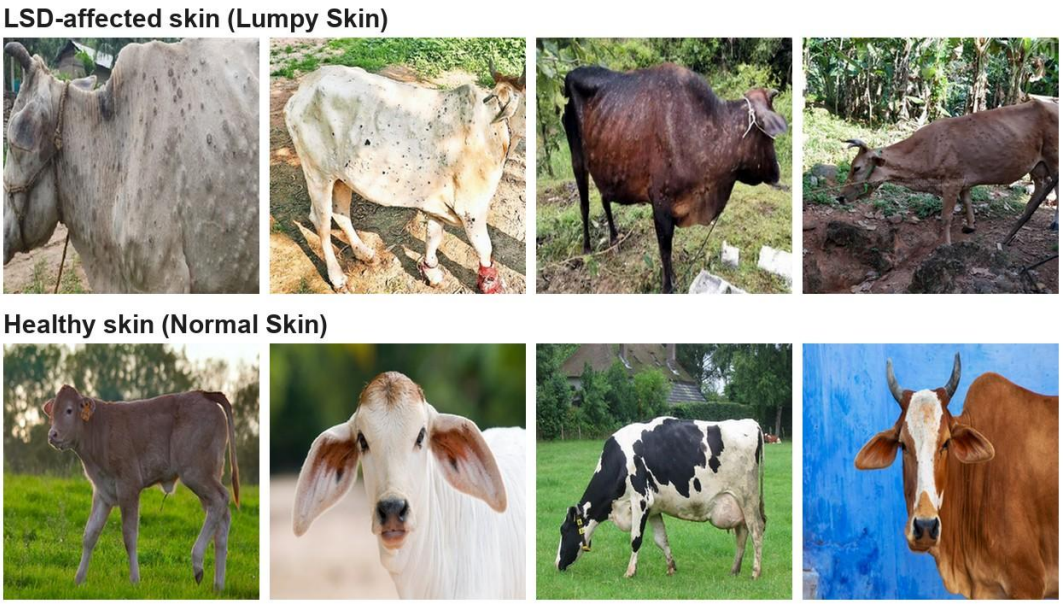


Figure 2.2: Dataset Samples: LSD-affected skin (top) and healthy skin (bottom)

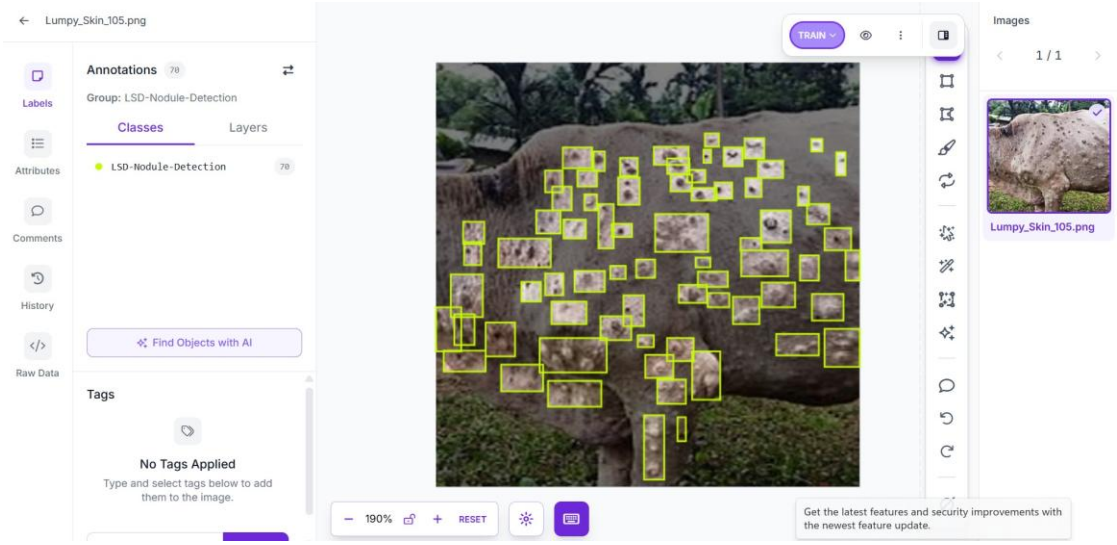


Figure 2.3: Example of manual nodule annotation with bounding boxes in Roboflow

2.1.3 Image Preprocessing

Photographs taken on farms vary enormously in resolution, lighting, viewing angles, background and distance from the animal. Preprocessing was therefore applied to give each model input in a consistent form, and augmentation was used to help the models handle a broader range of conditions beyond those represented in the original images alone could provide. Augmentation was applied only to the training data, so that the validation and test results reflect performance on unaltered images.

For the detection dataset, preprocessing was applied in Roboflow before training. Each image was automatically oriented according to its EXIF data, resized to 640 by 640 pixels

with black padding so that the original aspect ratio was preserved, and enhanced with adaptive histogram equalization (auto-contrast). The contrast enhancement was intended to make faint nodules more clearly visible against the coat, which can be especially for light-colored animals and photographs taken in poor lighting.

At inference time, every uploaded image is first checked against the permitted formats (JPEG, PNG, BMP and WebP) and a maximum size of 10 MB. It is then decoded directly from memory using OpenCV, without being written to disk, and converted from OpenCV's default BGR colour order to RGB. For detection, the image is resized to 640 by 640 pixels, which is handled internally by the YOLOv8 framework. For classification, each cropped region, and the full image, is resized to 224 by 224 pixels and normalized using the standard ResNet50 preprocessing function, which converts the image to the colour ordering and mean-centered values used when the network was originally trained on ImageNet.

The two models were trained with different augmentation strategies, each suited to their task. For YOLOv8s, augmentation was applied at two levels. First, Roboflow generated three augmented versions of every training image using random horizontal and vertical flips, random 90-degree rotations and brightness adjustments of up to 15 percent in either direction, which expanded the training set from 182 to 546 images. Second, the built-in augmentation pipeline of the Ultralights framework was applied during training. Mosaic augmentation, which stitches four training images into one, is particularly useful for small-object detection because it presents nodules at many different scales and positions. It was disabled for the final ten epochs so that the model could settle on realistic, unstitched images. For ResNet50, augmentation was implemented in the TensorFlow data pipeline. The settings are provided in Table 2.2.

Table 2.2: Data augmentation applied to the training sets

Model	Augmentation technique	Setting
YOLOv8s (Roboflow, offline)	Horizontal and vertical flip	Probability 0.5 each
YOLOv8s (Roboflow, offline)	Rotation	None, 90 degrees clockwise, 90 degrees counterclockwise or 180 degrees, equal probability
YOLOv8s (Roboflow, offline)	Brightness adjustment	Between -15% and +15%; three versions per training image
YOLOv8s (Ultralytics, online)	Mosaic (four-image composition)	Probability 1.0, disabled for the last 10 epochs
YOLOv8s (Ultralytics, online)	Horizontal flip	Probability 0.5
YOLOv8s (Ultralytics, online)	HSV colour jitter	Hue 0.015, saturation 0.7, value 0.4
YOLOv8s (Ultralytics, online)	Translation and scaling	Translate 0.1, scale 0.5
ResNet50	Horizontal and vertical flip	Random, each with probability 0.5
ResNet50	Rotation	Random multiples of 90 degrees
ResNet50	Brightness adjustment	Random, maximum delta 0.15

These operations preserve the features that matter for diagnosis, namely the raised, rounded shape of the nodules and their contrast with the surrounding skin, while varying the orientation, framing and illumination under which they appear.

2.1.4 YOLOv8s-Based LSD Nodule Detection

The detection module identifies and localizes individual LSD nodules in a photograph. Localization matters for two reasons. First, it enables the classifier to concentrate on regions that are most likely to contain evidence of disease rather than on the background or healthy parts of the animal. Second, it lets the system show the user exactly which areas of the skin led to the result, which makes the output easier to trust and easier for a veterinarian to verify.

YOLOv8s, the small variant of the YOLOv8 family, was selected for this task. It provides a suitable trade-off between accuracy and speed, with around 11.1 million parameters and 28.4 GFLOPs, which is light enough to run on a CPU-based server while still being considerably more capable than the nano variant. Its anchor-free detection head is well suited to objects such as nodules, whose size and aspect ratio vary from image to image.

The model was initialized with pretrained weights from the COCO dataset and fine-tuned on the annotated LSD dataset using Google Colab with an NVIDIA Tesla T4 GPU. Because the dataset is modest in size, a relatively low initial learning rate and mild dropout were used to limit the risk of overfitting, and early stopping was applied so that

training would end once the validation mAP stopped improving. Table 2.3 presents the training configuration.

Table 2.3: YOLOv8s training configuration

Parameter	Value
Base model	YOLOv8s pretrained on COCO (yolov8s.pt)
Input image size	640 × 640 pixels
Batch size	16
Maximum epochs	150
Early stopping patience	30 epochs
Initial learning rate (lr0)	0.001
Optimizer	Selected automatically by the Ultralytics framework
Dropout	0.1
Number of classes	1 (LSD nodule)
Training hardware	Google Colab, NVIDIA Tesla T4 (15 GB)

During training, early stopping ended the run after 78 epochs, with the best validation performance recorded at epoch 48. The weights from that epoch were saved and used in the deployed system. Training took approximately 16 minutes.

At inference time, the detector is run with a deliberately low confidence threshold of 0.10 and an Intersection over Union (IoU) threshold of 0.45 for non-maximum suppression. The low threshold was chosen because, in a screening context, missing a real nodule is more harmful than reporting a doubtful one, particularly as every detection is checked again by the classifier in the next stage. If no regions are found at this threshold, but the classifier considers the whole image to be at least moderately suspicious (a probability of 0.30 or above), the detector runs a second time with an even more sensitive threshold of 0.06. This second pass helps in cases where nodules are small or faint.

Each detected nodule is drawn on a copy of the image as an orange bounding box labelled "Nodule". The line thickness and font size are scaled to the resolution of the image so that the annotations remain legible on both small and large photographs. Figure 2.4 shows the nodule regions returned by the deployed system for two images from the dataset.

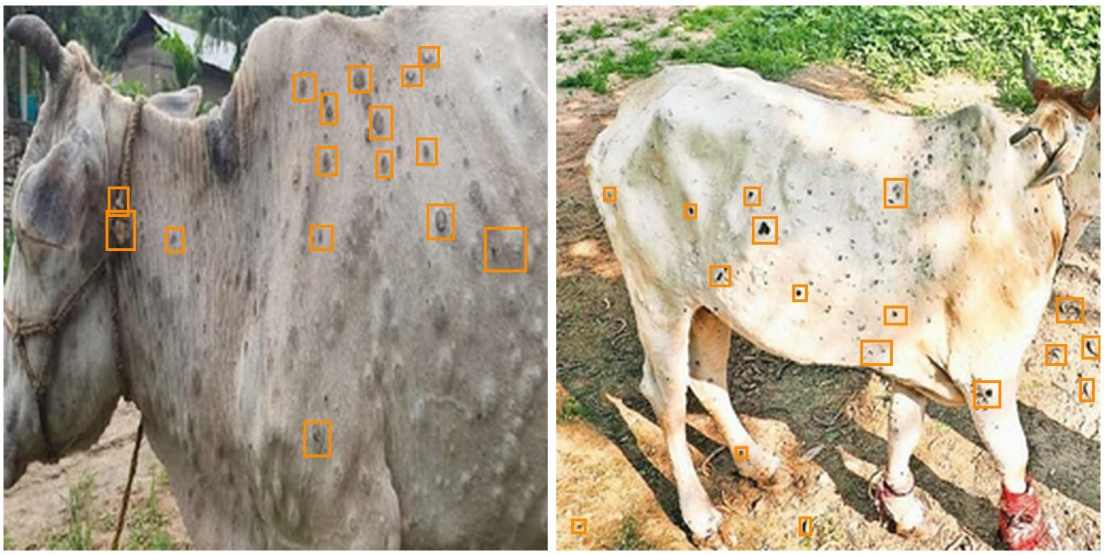


Figure 2.4: Detected nodule regions by the YOLOv8s pipeline on two LSD-affected images.

2.1.5 ResNet50-Based LSD Classification and Region Scoring

While the detector finds candidate nodules, it does not by itself decide whether the animal has LSD. Many things can resemble a nodule in a photograph, including insect bites, ticks, scars, dirt and patches of darker hair. The classification module therefore examines each detected region more closely and estimates the probability that it represents LSD.

ResNet50 was chosen for this stage because its residual connections allow a deep network to be trained reliably and because its ImageNet-pretrained features transfer well to new visual domains, which is important when only a few hundred labelled images are available. The original 1,000-class output layer was replaced with a compact classification head consisting of a global average pooling layer, a fully connected layer with 128 units and ReLU activation, a dropout layer with a rate of 0.3, and a single output unit with sigmoid activation. The sigmoid output gives a continuous probability between 0 and 1, rather than a hard label, which is what the later fusion and risk stages require. The resulting architecture shows in Figure 2.5.

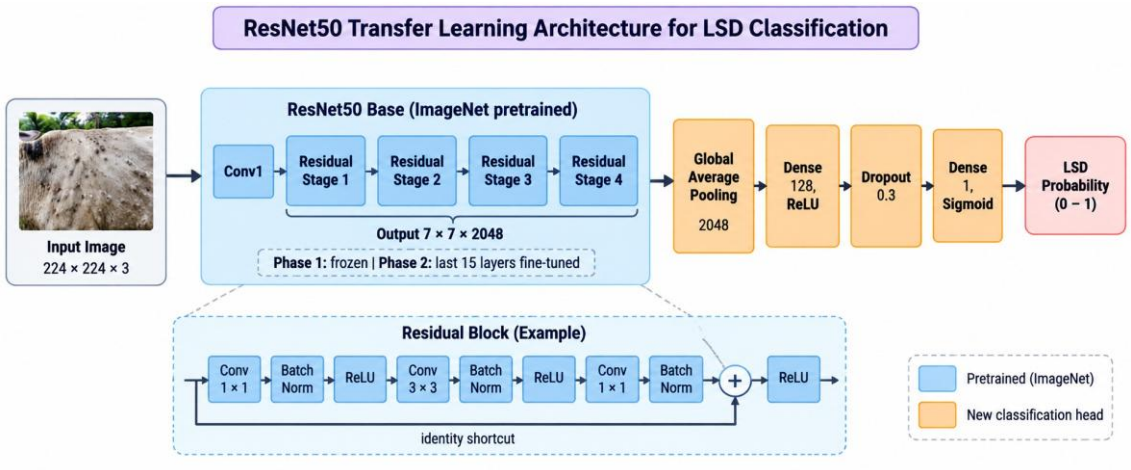


Figure 2.5: ResNet50 transfer learning architecture used for LSD classification.

The classifier was trained in two phases, a common strategy for transfer learning on small datasets. During the first paragraph, the entire ResNet50 base remained frozen while only the new classification head was trained, allowing the randomly initialized layers to adapt without affecting the pretrained features. During the second paragraph, the final fifteen layers of the base were unfrozen, and the network underwent further training with a much lower learning rate, allowing the deepest features to adjust gradually to the appearance of cattle skin. In both phases, the model with the highest validation AUC was saved. Table 2.4 presents the training configuration.

Table 2.4: ResNet50 training configuration

Parameter	Phase 1 (head training)	Phase 2 (fine-tuning)
Trainable layers	Classification head only (262,401 parameters)	Head and last 15 layers of ResNet50
Optimizer and learning rate	Adam, 1×10^{-3}	Adam, 1×10^{-5}
Maximum epochs (epochs run)	30 (27)	20 (16)
Loss function	Binary cross-entropy	Binary cross-entropy
Batch size	16	16
Callbacks	Checkpoint on best validation AUC, early stopping (patience 10), learning rate reduction on plateau (factor 0.5, patience 5)	Checkpoint on best validation AUC, early stopping (patience 10)
Input size	224 × 224 × 3	224 × 224 × 3

Once the two models are combined, the classifier is used in two ways. Every region returned by the detector is cropped from the original image and classified, giving a classification probability for that region. A region score is then calculated by multiplying the detector's confidence by the classifier's probability:

$$region\ score = detection\ confidence \times classification\ probability$$

Multiplying the two values means that a region scores highly only when both models agree that it is likely to be a nodule caused by LSD. A region that the detector is unsure about, or that the classifier considers to be healthy skin, is naturally given less weight. In addition, the classifier evaluates the complete image, which provides a safeguard when the detector misses nodules that are small or diffuse. The image-based probability is taken as the larger of the highest region score and the whole-image probability:

$$P_{image} = max\ (highest\ region\ score,\ whole\text{-}image\ probability)$$

If no regions are detected, the whole-image probability is used directly. The overall flow of the integrated detection and classification pipeline is shown in Figure 2.6.

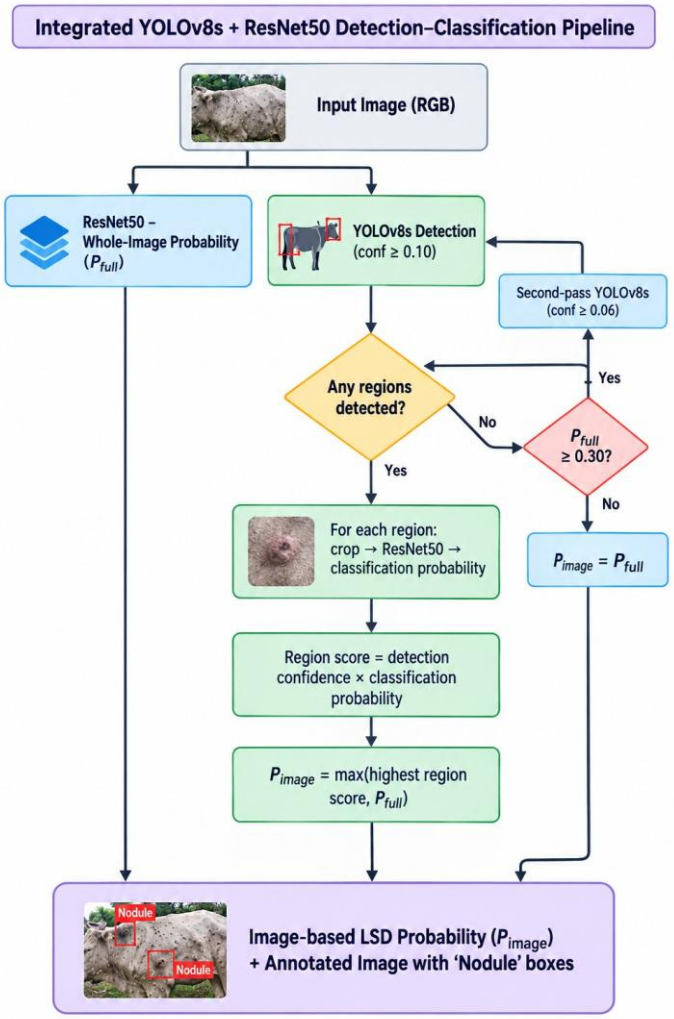


Figure 2.6: Integrated YOLOv8s and ResNet50 detection-classification pipeline

2.1.6 Clinical Symptom-Based Assessment

A photograph captures only part of the clinical picture. LSD is a systemic disease, and some of its most telling signs, such as swollen superficial lymph nodes and high fever, are invisible to a camera and may appear before, or alongside, the first nodules. To make use of this information, the component includes an optional clinical symptom checklist that the farmer can complete when submitting an image.

The checklist contains six yes-or-no questions and one numerical field for body temperature. Each sign is assigned a weight reflecting how specific it is to LSD, based on published clinical descriptions of the disease [1], [3]. Swollen lymph nodes and high fever receive the highest weights because they are among the most characteristic early indicators. Reduced milk yield, nasal discharge, ocular discharge and decreased appetite are real but non-specific signs shared with many other cattle illnesses and so receive smaller weights. Nasal and ocular discharge are scored separately because an animal may show one without the other. Table 2.5 presents the weights.

Table 2.5: Clinical symptom weights used in the LSD symptom assessment.

Clinical sign	Input type	Weight	Clinical interpretation
Swollen lymph nodes	Yes / No	2.5	Lymphadenitis, one of the most specific early indicators of LSD
High fever	Yes / No	2.0	Prodromal sign that can precede visible nodules by one to two days
Body temperature	Numerical (°C)	1.5	Scaled linearly from 0 at 38.5 °C (normal) to full weight at 41.5 °C (severe fever)
Reduced milk production	Yes / No	1.0	Common systemic effect, not specific to LSD
Nasal discharge	Yes / No	0.75	Non-specific respiratory sign
Eye discharge	Yes / No	0.75	Non-specific ocular sign
Decreased appetite	Yes / No	0.75	Non-specific systemic sign

The symptom probability is calculated as the sum of the weights of the signs that are present, divided by the total possible weight. For body temperature, the contribution is proportional to how far the reading lies between the normal and severe values and is limited to the range from zero to the full weight. The symptom probability is therefore a value between 0 and 1:

$$P_{symptom} = \Sigma\,(w_i \times s_i) \,/\, \Sigma\, w_i$$

where w_i is the weight of sign i and s_i is 1 if the sign is present and 0 if it is absent (or the scaled value in the case of body temperature). If the farmer leaves every box unchecked and does not enter a temperature, the symptom branch returns no result at all rather than a

probability of zero. This distinction matters, because an empty checklist means that no information was provided, not that the animal is healthy, and treating it as zero would unfairly pull down the final score.

As noted earlier, the checklist intentionally excludes questions about the presence, number or location of nodules. The image branch already measures these directly from the photograph, and a farmer's estimate of the same thing would add a second, less reliable copy of the same evidence. Restricting the checklist to signs the camera cannot see ensures that the two branches contribute genuinely independent information.

2.1.7 Hybrid Decision-Making Framework

The hybrid decision-making framework combines the outputs of the two branches into a single, interpretable result. The image-based probability and the symptom-based probability are fused using a weighted average:

$$P_{final} = 0.7 \times P_{image} + 0.3 \times P_{symptom}$$

The image branch is given greater weight because it provides direct visual evidence of the lesions that define the disease and is based on trained models, whereas the symptom branch relies on the farmer's observations, which are valuable but less precise. The 70:30 ratio is also consistent with the weighting used in the Mastitis module of CattleSense, which keeps the platform's behavior uniform across modules. When the farmer does not provide any symptoms, the symptom weight is set to zero and the image-based probability becomes the result, so the system degrades gracefully to an image-only assessment.

The final probability is mapped to one of three risk levels, each accompanied by specific guidance, as shown in Table 2.6. For a binary decision, a final probability of 0.30 or above is treated as LSD-positive. This threshold was chosen to coincide with the boundary between Low and Moderate risk, so that any animal for which the system recommends isolation or veterinary attention is also counted as a positive case. The lower-than-usual threshold reflects the priority of a screening tool, which is to avoid missing infected animals in a highly contagious disease.

Table 2.6: Risk levels and guidance generated by the system.

Final probability	Risk level	Guidance provided to the farmer
Below 30%	Low risk	Continue monitoring. Maintain regular health checks.
30% to below 70%	Moderate risk	Isolate the animal and monitor closely. Consider consulting a veterinarian.
70% and above	High risk	Immediate veterinary consultation strongly advised. Isolate the animal from the herd.

The framework also includes fallback behavior for robustness. If one of the two models cannot be loaded, the pipeline continues with the other. If image analysis fails entirely, the service returns a result based on the reported symptoms alone and indicates clearly that this is what has happened. Figure 2.7 illustrates the complete decision process.

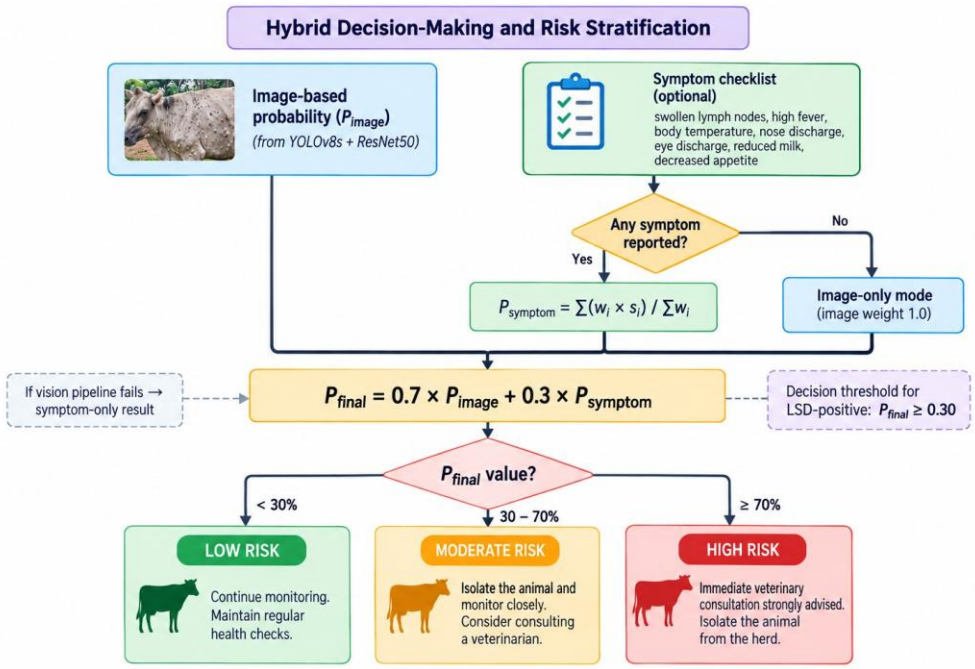


Figure 2.7: Hybrid decision-making and risk stratification process

The final output returned to the user includes the following:

- Final classification (LSD positive or healthy) and overall LSD probability
- Risk level (Low, Moderate or High) with corresponding guidance.
- Image-based probability, number of detected nodules, and the detection confidence, classification probability and region score of each nodule.
- Symptom-based probability, symptom risk label and the individual signs reported.
- The weights applied to the image and symptom branches and the sources used.
- An annotated image with every detected nodule marked.
- A downloadable PDF report summarizing the assessment.

2.1.8 Web-Based System Implementation

The LSD component was implemented as part of the CattleSense web platform so that farmers and veterinary officers can use it through an ordinary web browser on a phone or computer. The implementation follows a microservice architecture with three main layers: a React frontend, a Flask backend gateway, and the LSD inference microservice. The deployment architecture is shown in Figure 2.8.

The **LSD microservice** is a Flask application that loads the YOLOv8s and ResNet50 models into memory when it starts, so that each request can be processed without reloading them. The inference logic is organized into separate modules for the vision pipeline, the symptom assessment, the fusion and the PDF report, which keeps each part independently testable and maintainable. Because the trained model files exceed the size limit for files in a GitHub repository, they are stored as split parts and automatically reassembled when the service starts. The service is packaged as a Docker image based on Python 3.11, with a CPU-only build of PyTorch to keep the image small and includes a health check that reports whether the models are loaded. The REST endpoints exposed by the service are shown in Table 2.7.

Table 2.7: REST API endpoints of the LSD microservice

Method	Endpoint	Description
GET	/health	Reports service status and whether the YOLOv8s and ResNet50 models are loaded
POST	/api/predict/image	Image-only analysis using the YOLOv8s and ResNet50 pipeline
POST	/api/predict/assisted	Image analysis combined with the optional symptom checklist through hybrid fusion (main endpoint used by the platform)
POST	/api/report/pdf	Generates a PDF report from a previously returned result, without repeating the analysis
GET	/api/info	Describes the API, its endpoints and the model configuration

The **backend gateway**, also built with Flask, is the single-entry point for the frontend. It authenticates users with JSON Web Tokens (JWT), forwards LSD requests to the microservice through the routes /api/modules/lumpy/predict-assisted and /api/modules/lumpy/report-pdf and saves assessment results to the animal's record in a cloud-hosted PostgreSQL database through the /api/modules/lumpy/assessments route. Each saved result is linked to the user and the specific animal, so a farmer can build up a history of assessments for every cow.

The frontend was developed with React 18, Vite and Tailwind CSS. The LSD detection page allows the user to select an animal from their records, upload a photograph or capture one directly with the device camera, and optionally complete the symptom checklist. After analysis, the result card displays the overall probability, the risk level and guidance, the annotated image, and a breakdown showing how the image and the symptoms each contributed to the decision. From the same screen the user can download the assessment report as an English or Sinhala PDF or save the result to the animal's record. The interface is available in both English and Sinhala, which makes it accessible to a wider range of Sri Lankan farmers.

The PDF report is generated on the server with the fpdf2 library. It is a one-page summary containing the annotated image, a probability gauge, a colour-coded risk badge, the guidance message, the breakdown of the assessment and a timestamp. The report can be downloaded in English or Sinhala and can be produced either from a fresh result or from a saved record, so a farmer can share it with a veterinary officer at any time. Screenshots of the implemented interface are shown in Figures 2.9 to 2.11.

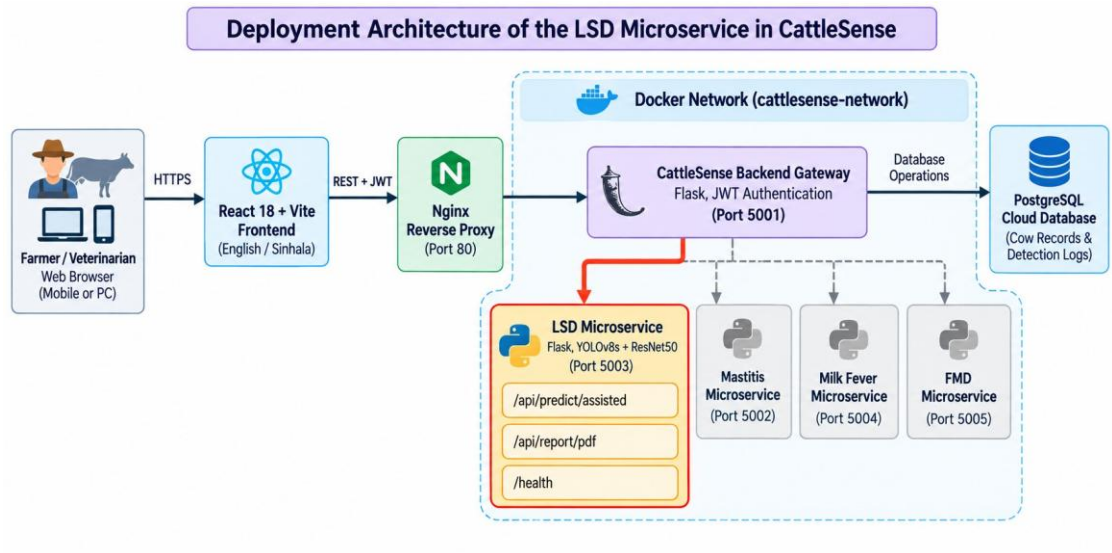


Figure 2.8: Deployment architecture of the LSD microservice within the CattleSense platform

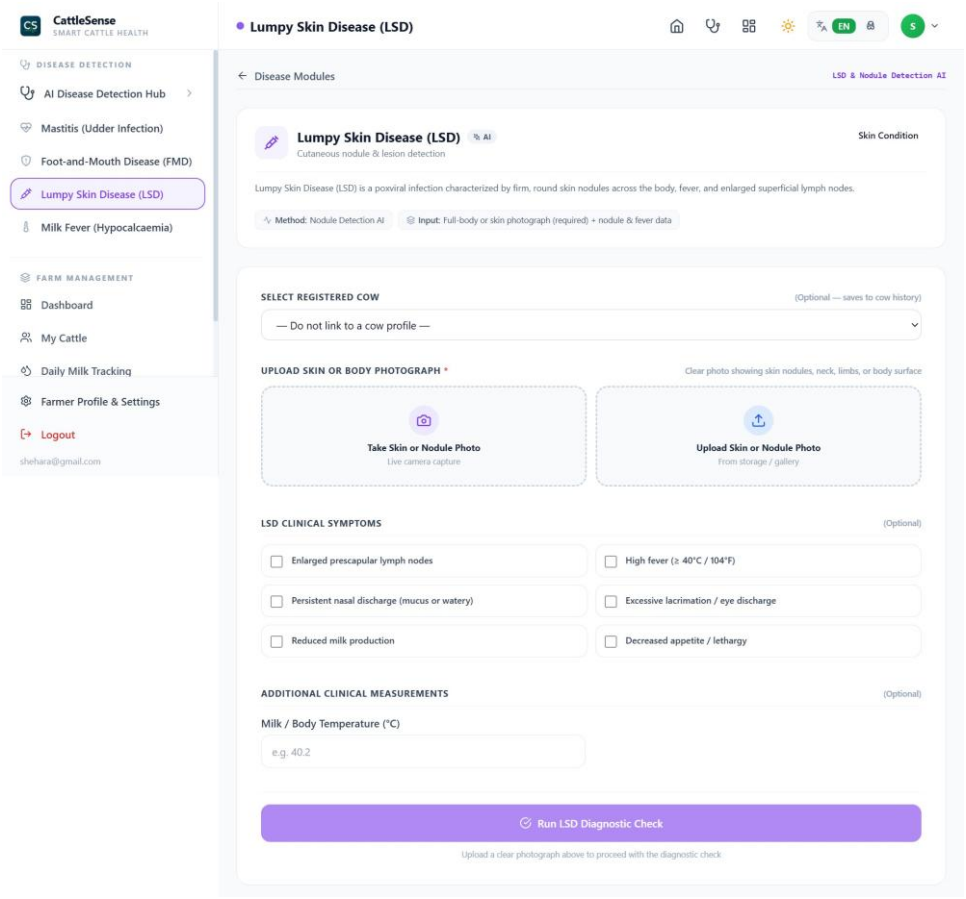


Figure 2.9: LSD detection page with image upload, camera capture and the optional symptom checklist

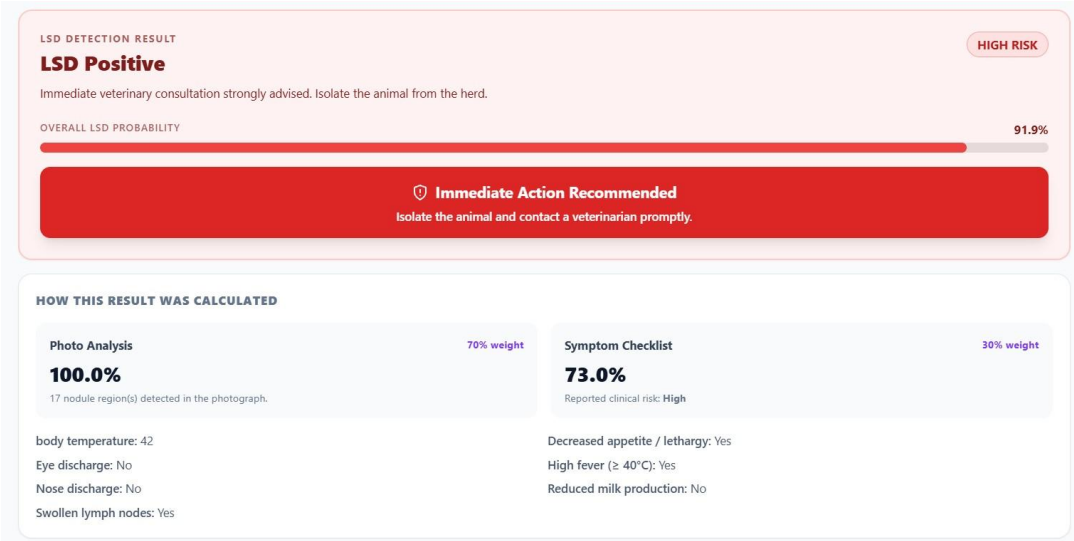


Figure 2.10: LSD result card showing probability, risk level, guidance and contribution

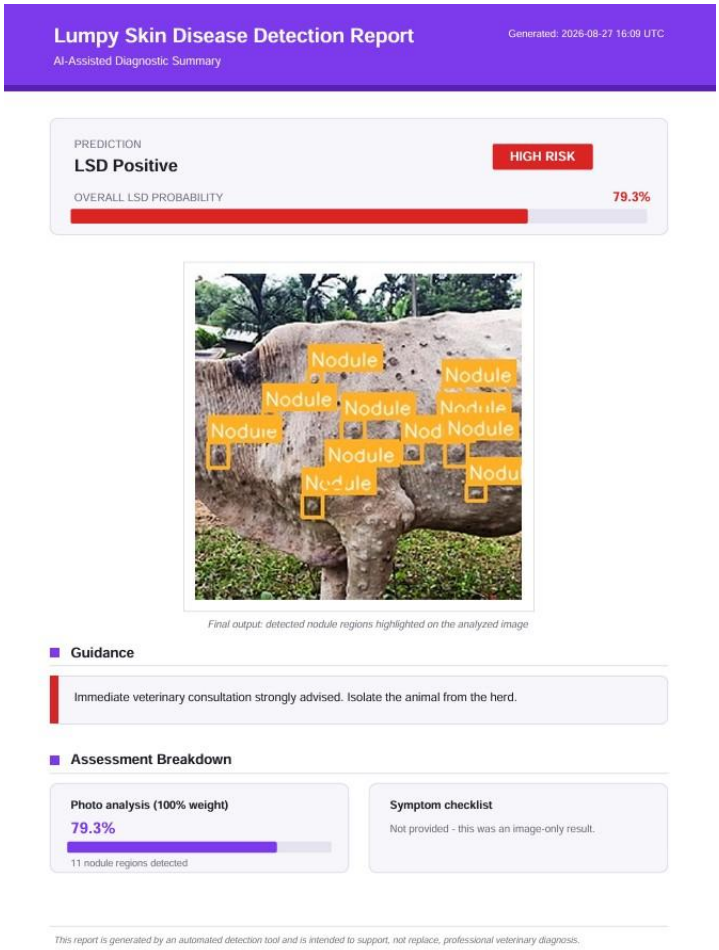


Figure 2.11: Generated PDF report with annotated nodule detection for veterinary reference

2.2 Testing and Validation

Testing and validation were carried out to assess the accuracy, reliability and robustness of each part of the LSD component as well as the system. Since the component combines two deep learning models, a rule-based scoring module and a web service, several complementary forms of evaluation were needed. The testing process focused on the following:

- Nodule detection performance of the YOLOv8s model
- Classification performance of the ResNet50 model
- End-to-end performance of the integrated detection and classification pipeline
- Correctness of the symptom scoring, hybrid fusion and risk stratification logic
- Functional behavior and error handling of the REST API
- Processing time per image

2.2.1 YOLOv8s Model Validation

The YOLOv8s detector was monitored on the validation set throughout training, and the weights corresponding to the best validation performance were retained. The final model was then evaluated once on the held-out test set of 24 images, which had not been used for training or model selection. Standard object detection metrics were used, namely precision, recall, F1-score, mAP@0.5 and mAP@0.5:0.95.

Precision indicates the proportion of regions predicted as nodules that were nodules, while recall indicates the proportion of actual nodules that the model successfully detected. The F1-scope provides a combined measure of these two metrics.mAP@0.5 summarizes localization quality across all confidence levels, counting a prediction as correct when it overlaps an annotated nodule by at least 50%. mAP@0.5:0.95 applies a stricter series of overlap thresholds. The box, classification and distribution of the confidence scores were also monitored to determine whether the model was learning steadily or beginning to overfit. In addition, the predictions on individual validation images were inspected visually to understand where false positives and missed nodules occurred.

2.2.2 ResNet50 Model Validation

The ResNet50 classifier was validated on the separate validation set during both training phases, with the checkpoint achieving the highest validation AUC retained. Its final performance was assessed using the held-out test set of 58 images. The metrics used were accuracy, precision, recall, F1-score and the area under the Receiver Operating Characteristic curve (ROC-AUC), together with the confusion matrix.

Because the dataset is imbalanced, accuracy alone could give a misleading picture: a model that labelled every image as LSD-positive would still appear fairly accurate. Precision, recall and F1-score were therefore reported separately for the healthy and LSD-positive classes, and the confusion matrix was examined to identify the types of errors made. ROC-AUC was used to evaluate how well the model's probability output separates the two classes across all possible decision thresholds, which is particularly relevant here because the system relies on the probability itself rather than a fixed label.

2.2.3 Hybrid Framework Validation

The integrated pipeline was evaluated end to end on the 24 images of the detection test set, of which 18 were LSD-positive and 6 were healthy. Each image was processed by the complete detection and classification pipeline, and the predicted outcome, using the 0.30 decision threshold, was compared with the known label. The evaluation recorded overall accuracy, false positive rate, false negative rate and average processing time per image. The false negative rate is of particular interest, because a missed case of a contagious disease carries the greatest risk.

The symptom assessment and fusion logic were verified by submitting requests with known inputs and checking that the returned probabilities matched the values expected from the formulas in Sections 2.1.6 and 2.1.7. Functional testing of the deployed microservice was also carried out by sending requests directly to its REST API, covering both normal use and invalid input. Table 2.8 presents the test cases and their outcomes.

Table 2.8: Functional test cases for the LSD microservice

ID	Test scenario	Expected result	Actual result	Status
TC01	Health check request	Service reports status and model availability	Status 'ok', both models loaded	Pass
TC02	Valid image, image-only endpoint	Image probability, detections and risk level returned	Result returned with image weight 1.0	Pass
TC03	Valid image with symptoms (swollen lymph nodes, high fever, 40.5 °C)	Symptoms scored and fused at 70:30	Symptom probability 0.595, fused probability 0.729, matching the formula	Pass
TC04	Valid image, no symptoms reported	Falls back to image-only result	Image weight 1.0, symptom weight 0.0	Pass
TC05	Checklist submitted with all signs unchecked	Treated as no symptom information	Symptom branch ignored, image-only result	Pass
TC06	Body temperature of 38.5 °C only	Temperature contributes zero	Symptom probability 0.0	Pass
TC07	Body temperature of 43 °C only	Temperature contribution capped at full weight	Symptom probability 0.162 (1.5 / 9.25)	Pass
TC08	Request without an image	HTTP 400 with clear error message	HTTP 400, 'Missing required field: image'	Pass
TC09	Unsupported file type (.gif)	HTTP 400, invalid format	HTTP 400, list of allowed formats returned	Pass
TC10	Empty image file	HTTP 400	HTTP 400, 'Uploaded image file is empty'	Pass
TC11	Corrupted or non-image file	HTTP 400	HTTP 400, 'Could not decode image file'	Pass
TC12	Malformed symptom data	HTTP 400	HTTP 400, 'symptoms must be valid JSON'	Pass
TC13	File larger than 10 MB	Request rejected	HTTP 413, request entity too large	Pass

ID	Test scenario	Expected result	Actual result	Status
TC14	PDF report from a full analysis result	PDF document returned	PDF returned (HTTP 200)	Pass
TC15	PDF report from a minimal saved record	PDF generated from available fields	PDF returned (HTTP 200)	Pass
TC16	PDF request without result data	HTTP 400	HTTP 400, 'Missing result data'	Pass

2.2.4 Performance Evaluation Metrics

The metrics used to evaluate the component were selected to suit each stage of the pipeline and are defined below, where TP, FP, TN and FN denote true positives, false positives, true negatives and false negatives respectively.

Table 2.9: Performance evaluation metrics

Metric	Definition	Applied to
Precision	TP / (TP + FP)	YOLOv8s, ResNet50, pipeline
Recall (sensitivity)	TP / (TP + FN)	YOLOv8s, ResNet50, pipeline
F1-score	2 × Precision × Recall / (Precision + Recall)	YOLOv8s, ResNet50, pipeline
Accuracy	(TP + TN) / (TP + TN + FP + FN)	ResNet50, pipeline
mAP@0.5	Mean average precision at an IoU threshold of 0.5	YOLOv8s
mAP@0.5:0.95	Mean average precision averaged over IoU thresholds from 0.5 to 0.95	YOLOv8s
ROC-AUC	Area under the curve of true positive rate against false positive rate	ResNet50
False positive rate	FP / (FP + TN)	Pipeline
False negative rate	FN / (FN + TP)	Pipeline
Processing time	Average time to analyses one image	Pipeline, deployed service

Together, these metrics allow the component to be judged on how well it localizes nodules, how reliably it separates infected from healthy animals, how often it makes the more dangerous kind of error, and whether it responds quickly enough to be practical.

2.3 Project Requirements

The requirements of the LSD component were defined at the proposal stage and refined during development. They cover the functions the system must perform, the qualities it must have, the needs of its users, and the hardware and software needed to build and run it.

2.3.1 Functional Requirements

- **Image input** - Allow the user to upload a cattle skin photograph or capture one directly with the device camera.
- **Input validation** - Accept only supported image formats (JPEG, PNG, BMP, WebP) up to 10 MB and reject empty, corrupted or invalid files with a clear message.
- **Nodule detection** - Detect and localize LSD nodules in the image using the YOLOv8s model.
- **LSD classification** - Estimate the probability of LSD for each detected region and for the whole image using the ResNet50 model.
- **Region scoring** - Combine detection confidence and classification probability into an image-based LSD probability.
- **Symptom assessment** - Allow the user to report clinical signs optionally and convert them into a weighted symptom probability.
- **Hybrid fusion** - Combine the image-based and symptom-based probabilities (70% image, 30% symptoms), using the image alone when no symptoms are reported.
- **Risk guidance** - Classify the final probability as Low, Moderate or High risk and display the corresponding guidance.
- **Visual evidence** - Display an annotated image with every detected nodule marked.
- Report generation - Generate a downloadable assessment report in English or Sinhala PDF format.
- **Record keeping** - Save each assessment to the corresponding animal's record for later review.
- **Fallback operation** - Continue to operate with one model, or with symptoms alone, if part of the vision pipeline is unavailable.

2.3.2 Non-Functional Requirements

- **Accuracy** - Identify LSD-affected cattle reliably, giving priority to minimizing missed cases (false negatives).
- **Performance** - Return results within a time that is acceptable for interactive use; the proposal sets a target of 10 to 15 seconds per image.
- **Usability** - Provide a simple interface that farmers with little technical knowledge can use, in both English and Sinhala.

- **Interpretability** - Present results as probability, a risk level, clear guidance and an annotated image rather than an unexplained label.
- **Reliability** - Handle invalid input and partial model failures gracefully without crashing the service.
- **Security and privacy** - Restrict access to authenticated users and protect stored records.
- **Scalability and maintainability** - Run as an independent, containerized microservice that can be updated, retrained or scaled without affecting the other CattleSense modules.
- **Portability** - Run on standard CPU-based servers without requiring a GPU.

2.3.3 User Requirements

The primary users of the component are cattle farmers, particularly smallholder dairy farmers, together with veterinary surgeons and field officers who support them. Their requirements are as follows:

- Upload or capture a photograph of an animal easily from a mobile phone or computer.
- Optionally record clinical signs they have observed, without needing specialist knowledge.
- Receive a clear indication of the likelihood of LSD and the level of risk.
- Understand what to do next through practical guidance, such as isolating the animal or contacting a veterinarian.
- See which areas of the skin the system identified as nodules.
- Download a report that can be shown or sent to a veterinary officer.
- Keep a history of assessments for each animal.

2.3.4 System Requirements

Hardware requirements

- Smartphone or digital camera for capturing images of cattle.
- Computer or smartphone with a modern web browser for accessing the platform.
- GPU-supported environment for model training (an NVIDIA Tesla T4 on Google Colab was used)
- Server or virtual machine with at least 4 GB of RAM (8 GB recommended) for CPU-based inference.

- Storage for the trained models (approximately 22.5 MB for YOLOv8s and 90 MB for ResNet50) and the datasets

Software requirements

- Operating system: Windows 10/11 or Ubuntu 20.04 or later
- Programming language: Python 3.11
- Deep learning frameworks: Ultralytics YOLOv8 (with PyTorch) and TensorFlow/Keras
- Supporting libraries: OpenCV, NumPy, Pillow, scikit-learn, Matplotlib and fpdf2.
- Backend: Flask with Flask-CORS and Flask-JWT-Extended
- Frontend: React 18, Vite, Tailwind CSS and Axios
- Database: PostgreSQL
- Deployment: Docker, Docker Compose and Nginx
- Development and training tools: Google Colab, Roboflow, Git and GitHub

2.4 Ethical and Security Considerations

Although the LSD component analyses images of animals rather than people, it still handles information that belongs to farmers and affects decisions about animal health and welfare. Ethical, social and security considerations were therefore taken into account throughout their development.

Above all, the component was designed as a screening and decision-support tool, not as a replacement for veterinary diagnosis. LSD is a notifiable disease in Sri Lanka, and confirmation and official reporting remain the responsibility of qualified veterinary officers. The system's guidance reflects this: whenever the risk is moderate or high, it advises the farmer to isolate the animal and seek veterinary attention.

2.4.1 Farmer and Animal Data Privacy

The images used to train the models show cattle and their surroundings and do not contain personal information about farmers. Within the platform, however, each assessment is linked to a user account and to a specific animal, and the cow records may contain details such as names, tag numbers and breeds. This information is stored only for the purpose of providing the service to the farmer who created it. Uploaded images are processed in memory and are not written on the server's disk by the LSD service; they are

kept only when the user chooses to save an assessment to the animal's record. Each user can see only their own animals and assessments.

2.4.2 Ethical Considerations

The training images were used solely for academic research and model development. The annotated detection dataset was managed through Roboflow under a Creative Commons Attribution (CC BY 4.0) license, and any third-party images are used in accordance with their license terms. A specific ethical concern for a disease-screening tool is the possibility of incorrect results. A false negative may give a farmer false reassurance and delay the isolation of an infected animal, while a false positive may cause unnecessary worry or cost. The system addresses this by favoring sensitivity in its decision threshold, by showing the evidence behind each result, and by presenting its output as guidance that encourages veterinary consultation rather than as a final diagnosis. The limitations of the models are also stated openly in this report.

2.4.3 Security Considerations

- **Authentication** - Access to the platform's detection and record features requires a valid JSON Web Token issued at login.
- **Network isolation** - The LSD microservice is not exposed directly to the frontend; it is reached only through the backend gateway over a private Docker network.
- **Input validation** - File type, size (10 MB limit) and content are checked before processing, protecting the service against malformed or oversized uploads.
- **Secure storage** - Assessment records are stored in a managed PostgreSQL database accessed over an encrypted connection, with credentials held in environment variables rather than in the source code.
- **Error handling** - Errors are returned as structured messages without exposing internal details of the server to the user.

Further measures, such as rate limiting and role-based access for veterinary officers, can be added to the modular architecture as the platform moves towards wider deployment.

2.5 Commercialization Aspects of the Product

2.5.1 Target Audience

The LSD detection component is aimed primarily at those who are directly responsible for the health of cattle in Sri Lanka and who are most affected when the disease is detected late. Its main target users are:

- Smallholder and medium-scale dairy and cattle farmers
- Large commercial dairy farms with sizeable herds
- Government veterinary surgeons and livestock field officers
- Private veterinary practitioners
- Dairy cooperatives and milk collection networks
- Livestock development organizations and NGOs working with rural farmers.

The component is especially valuable in rural areas where veterinary services are stretched, and farmers may have to wait several days for a professional visit. Before it is used commercially in a clinical capacity, it would need further validation on larger, locally collected datasets and the endorsement of the relevant veterinary authorities.

2.5.2 Market Space

Sri Lanka's cattle population of around 1.2 million animals is spread across a very large number of small farms, most of which have no access to diagnostic technology of any kind [5]. The repeated LSD outbreaks in the region have shown how quickly the disease can spread and how costly delayed detection can be, which creates a clear need for affordable, accessible screening tools. At present, no widely available digital tool offers LSD screening to Sri Lankan farmers, leaving the market largely open.

The component has several characteristics that strengthen its market position. It works with an ordinary smartphone photograph and requires no special equipment. It runs in a web browser, so there is nothing to install. It supports Sinhala as well as English. Most importantly, it is part of CattleSense, which addresses four major cattle diseases in one platform and maintains a shared record for each animal. A farmer or veterinary office is far more likely to adopt one platform covering several diseases than separate tools for each. Beyond Sri Lanka, the same approach could be adapted for other South and Southeast Asian countries where LSD has recently become established.

2.5.3 Revenue Earning

Since many of the intended users are smallholder farmers with limited income, the proposed business model combines free access for individual farmers with paid services for organizations. The model is planned at the level of the CattleSense platform, with the LSD component forming one of its core features.

- **Freemium access for farmers** - Basic LSD screening free of charge, with a low-cost premium tier (approximately LKR 500 per month) offering unlimited assessments, full assessment history and PDF reports.
- **Institutional subscriptions** - Monthly subscriptions (approximately LKR 5,000 to 10,000) for veterinary practices, large farms and dairy cooperatives, including multi-herd management and reporting features.
- **Annual licenses** - Annual licenses for government veterinary offices, regional livestock departments and NGOs, supporting disease surveillance across their areas.
- **Partnerships** - Collaboration with dairy processing companies, livestock insurers and agricultural input suppliers who benefit from healthier herds and may sponsor access for their supplier farmers.

The web-based, containerized deployment keeps operating costs low and allows updates, such as retrained models, to be rolled out centrally to every user at once.

2.6 Tools and Technologies

A range of tools and technologies were used to develop the LSD component, covering data preparation, model training, inference, web development and deployment. Each was selected for a specific reason, as summarized in Table 2.10.

Table 2.10: Tools and technologies used, with justification.

Tool / technology	Purpose in this component	Reason for selection
Python 3.11	Main programming language for training, inference and the microservice	Rich ecosystem of machine learning libraries and direct support from all frameworks used
Ultralytics YOLOv8 (PyTorch)	Training and running the YOLOv8s nodule detector	State-of-the-art real-time detector with a simple training interface and pretrained weights
TensorFlow / Keras	Building, training and running the ResNet50 classifier	Provides ImageNet-pretrained ResNet50 and a convenient API for two-phase transfer learning
OpenCV	Image decoding, colour conversion, cropping and drawing annotations	Fast and reliable image processing that works directly with in-memory images
NumPy, scikit-learn, Matplotlib	Numerical processing, data splitting, metrics and plots	Standard, well-tested libraries for data handling and evaluation
Roboflow	Annotation, augmentation and versioning of the detection dataset	Purpose-built annotation tool with direct export to YOLO format
Google Colab (NVIDIA Tesla T4)	Model training	Free GPU access sufficient for training both models
Flask and Flask-CORS	LSD microservice and backend gateway REST APIs	Lightweight Python framework, consistent with the other CattleSense services
fpdf2 and Pillow	Generating the downloadable PDF report	Lightweight PDF generation without external dependencies
React 18, Vite, Tailwind CSS, Axios	Frontend interface, camera capture and result display	Fast development, responsive layouts and simple API communication
Flask-JWT-Extended	Authentication of platform users	Standard token-based authentication for REST APIs
PostgreSQL (cloud-hosted)	Storing animal records and assessment results	Reliable relational database shared across all CattleSense modules
Docker, Docker Compose and Nginx	Containerization, orchestration and reverse proxy	Consistent deployment and isolation of each microservice
Git and GitHub	Version control and collaboration within the research group	Enables parallel development of the four modules

Together, these tools made it possible to build a complete pipeline, from annotating the data and training the models to delivering the results to farmers through a secure and accessible web platform.

3. RESULTS AND DISCUSSION

3.1 Results

This chapter presents the results obtained from the LSD detection component and discusses what they mean. The evaluation was carried out in stages. The YOLOv8s nodule detector and the ResNet50 classifier were first evaluated separately on their own held-out test sets. The integrated detection and classification pipeline was then evaluated end to end, first as the prototype version developed in the training notebooks and then as the version deployed in the CattleSense platform. Finally, the behavior of the hybrid fusion of image and symptom evidence was examined. All results reported here were obtained from the project's training notebooks or by running the deployed service; no values have been estimated.

Dataset Summary

Two datasets were used, as described in Section 2.1.2. The classification dataset contained 382 images, of which 320 showed LSD-affected cattle and 62 showed healthy cattle. It was divided by stratified sampling into 266 training, 58 validation and 58 test images, preserving a ratio of roughly five positive images to every healthy one in each subset. The detection dataset contained 254 original images divided into 182 training, 48 validation and 24 test images. The training images were expanded threefold by augmentation to 546. The distribution of both datasets is shown in Figure 3.1.

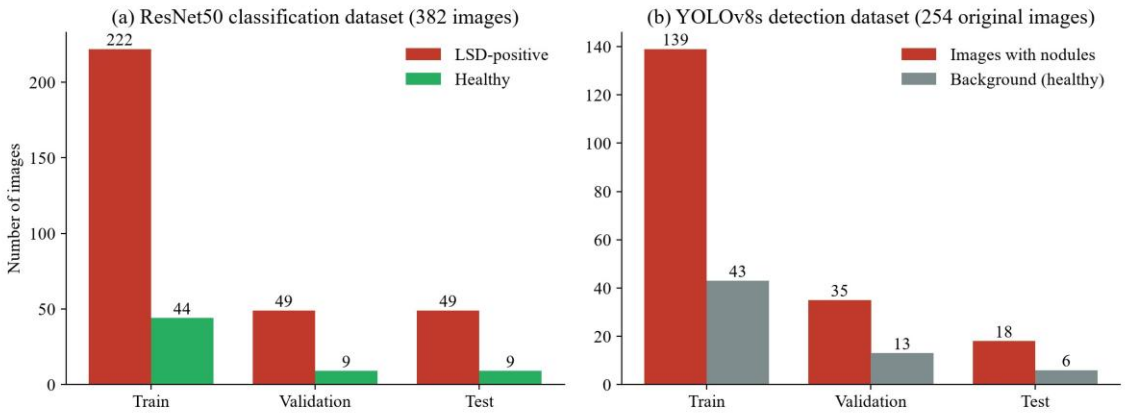


Figure 3.1: Class distribution across ResNet50 and YOLOv8s dataset splits

The detection dataset contained many small objects. Across the three subsets, 7,289 nodules were annotated: 6,528 in the augmented training set, 500 in the validation set and 261 in the test set. An image that contained nodules held 14 to 16 annotated nodules on

average, with a median of about 10 and a maximum of 84 in a single image. The nodules were also very small relative to the image: the median bounding box covered only about 6% of the image width and height, and roughly four in ten boxes were narrower than 5% of the image width. These statistics are summarized in Table 3.1.

Table 3.1: Annotation statistics of the YOLOv8s detection dataset

Statistic	Training (augmented)	Validation	Test
Images	546	48	24
Background (healthy) images	129	13	6
Annotated nodules	6,528	500	261
Mean nodules per positive image	15.7	14.3	14.5
Median nodules per positive image	11	9	10.5
Maximum nodules in one image	83	84	55
Median box width / height (fraction of image)	0.056 / 0.059	0.059 / 0.069	0.056 / 0.059
Boxes narrower than 5% of image width	39%	36%	41%

Sample annotated training images are shown in Figure 3.2. They illustrate the difficulty of the task: nodules appear on animals of many coat colors, often in dense clusters, and the adaptive contrast enhancement applied in Roboflow has noticeably altered the colour and texture of some images. The samples also reveal an important characteristic of the annotations, discussed further in Section 3.3.1: in several images only a portion of the visible nodules were labelled.



Figure 3.2: Sample annotated training images with nodule bounding boxes

3.1.1 YOLOv8s Detection Results

The YOLOv8s detector was trained for a maximum of 150 epochs with an early stopping patience of 30 epochs. Training stopped after 78 epochs because the validation mAP had not improved for 30 epochs, and the best weights, recorded at epoch 48, were retained. Training took 0.273 hours (about 16 minutes) on an NVIDIA Tesla T4 GPU. The results on the validation set at the best epoch and on the held-out test set are given in Table 3.2.

Table 3.2: YOLOv8s detection performance

Metric	Validation (best epoch 48)	Test (held out)
Precision	21.82%	19.05%
Recall	21.00%	27.59%
F1-score	21.40%	22.54%
mAP@0.5	13.83%	11.83%
mAP@0.5:0.95	3.45%	3.09%
Images / annotated nodules	48 / 500	24 / 261
Inference time per image (Tesla T4)	5.9 ms	16.3 ms
Epochs run / training time	78 epochs / 16.4 minutes	-

On the held-out test set the detector achieved a mAP@0.5 of 11.83%, a precision of 19.05% and a recall of 27.59%, giving an F1-score of 22.54%. These figures are well below the 80% mAP target set for the detector in the project proposal. The training and validation curves in Figures 3.3 and 3.4 help to explain why. The training box loss fell

steadily from 3.19 to about 2.19, and the training classification and distribution focal losses also decreased throughout, showing that the model continued to fit the training data. The validation box loss, however, reached its lowest point at around epoch 23 and then rose steadily to above 3.0 by the end of training, and the validation distribution focal loss followed the same pattern. In other words, after roughly 25 epochs the model began to memories the training images rather than learn features that generalize. The validation mAP rose until it peaked at 13.83% at epoch 48 and then fluctuated without further improvement, which is why early stopping ended training 30 epochs later.

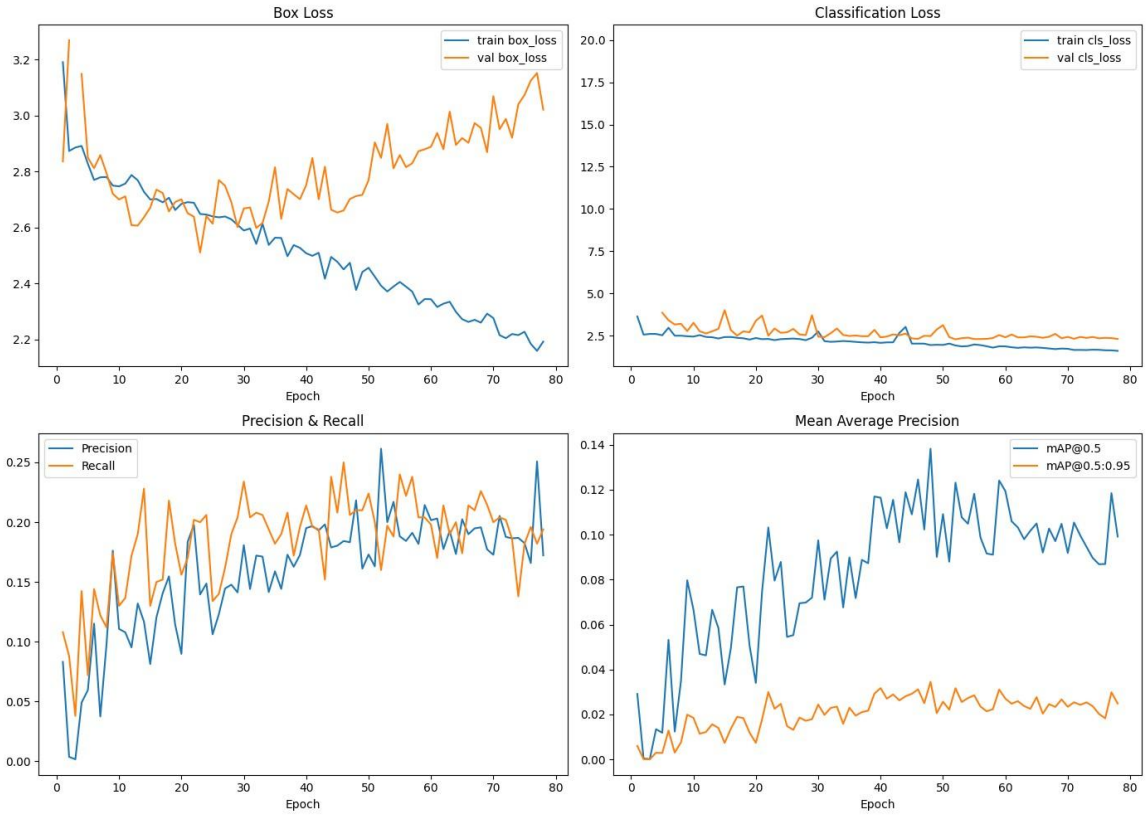


Figure 3.3: YOLOv8s training and validation performance across 78 epochs

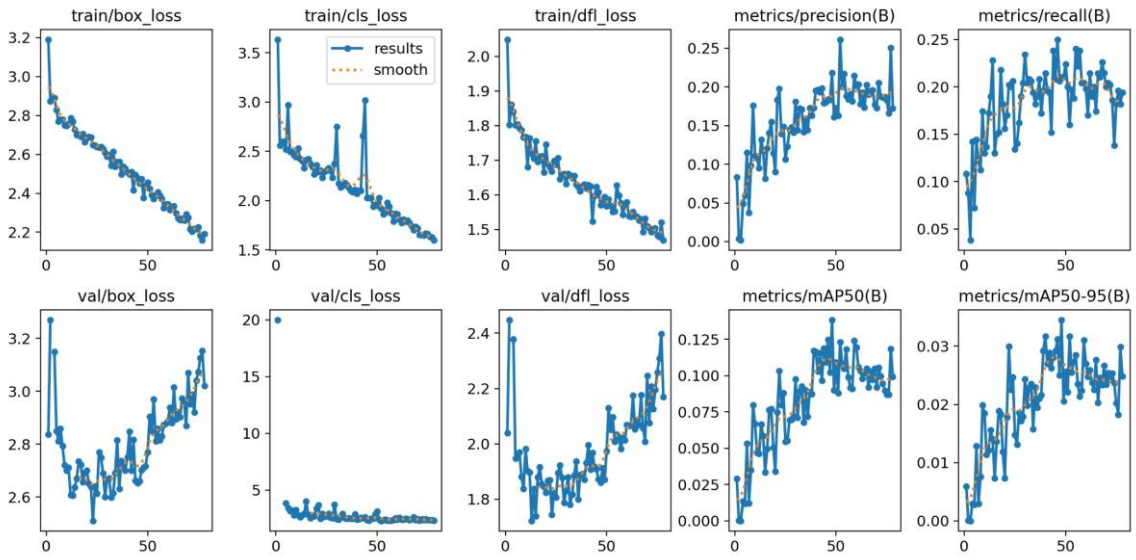


Figure 3.4: YOLOv8s full training dashboard generated by the Ultralytics framework.

The confusion matrix produced by the Ultralytics validator on the validation set (Figure 3.5) shows the same picture at the level of individual nodules. Of the 500 annotated nodules, 80 were correctly detected and 420 were missed, while 120 detections were made on regions that had not been annotated as nodules. The detector therefore missed the majority of the labelled nodules, although, as discussed in Section 3.3.1, part of what is counted here as error is a consequence of incomplete annotation rather than of the model alone.

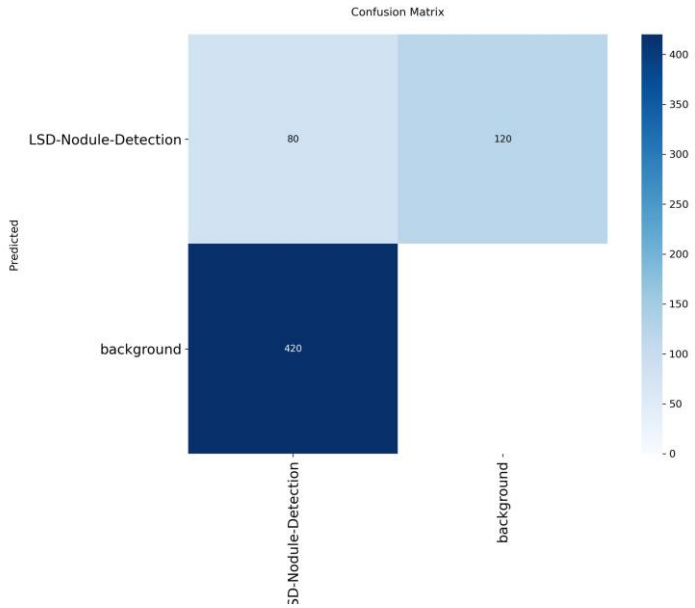


Figure 3.5: YOLOv8s confusion matrix on the validation set (nodule versus background)

Figure 3.6 shows the detector's predictions on six validation images at a confidence threshold of 0.35. The two healthy animals in the top row correctly produced no detections. On the LSD-affected animals, however, the behavior was inconsistent: the detector found seven nodules on one animal (bottom left) but returned no detections at all on three others, even though their lesions are clearly visible. This confirms that the detector, used on its own with a conventional threshold, misses too many nodules to serve as a reliable basis for a screening decision.

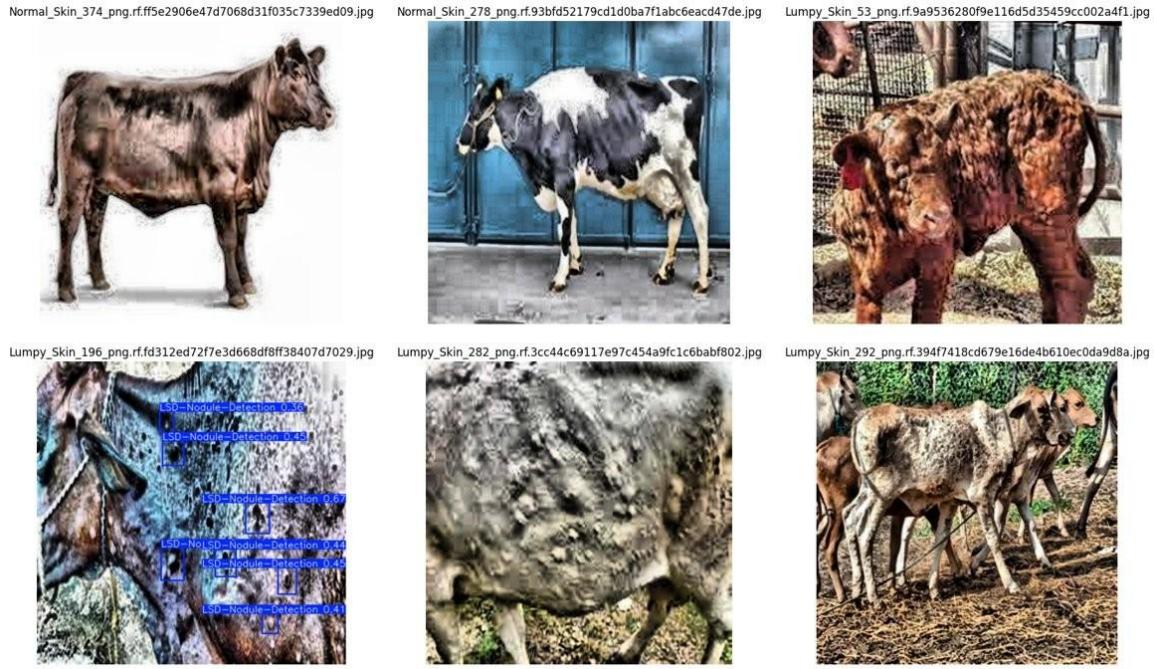


Figure 3.6: YOLOv8s predictions on validation images at a confidence threshold of 0.35

3.1.2 ResNet50 Image Classification Results

The ResNet50 classifier was trained in two phases, as described in Section 2.1.5: 27 epochs with the base network frozen, followed by 16 epochs of fine-tuning the last fifteen layers. The training curves are shown in Figure 3.7. Training accuracy rose quickly and reached 100% within the first phase, while validation accuracy settled at about 94.8%. The validation AUC remained high throughout, between approximately 0.98 and 0.99, and peaked at 0.992 during the first phase. After fine-tuning began, the validation loss stabilized at around 0.12 to 0.14, while the training loss approached zero. This gap indicates some overfitting to the training images, which is expected with a dataset of this size.

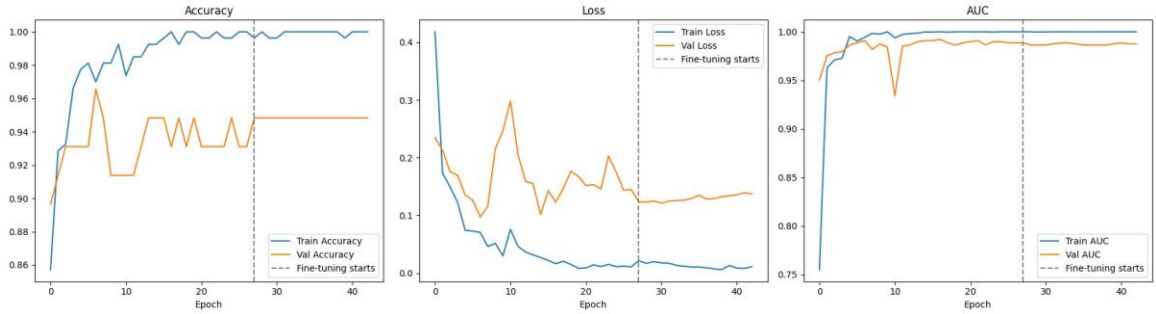


Figure 3.7: ResNet50 training curves: accuracy, loss and AUC across both training phases

The best model was evaluated once on the held-out test set of 58 images. The results are summarized in Table 3.3.

Table 3.3: ResNet50 classification results on the held-out test set (58 images)

Metric	Healthy (9 images)	LSD-positive (49 images)	Overall
Precision	0.86	0.94	0.90 (macro) / 0.93 (weighted)
Recall	0.67	0.98	0.82 (macro) / 0.93 (weighted)
F1-score	0.75	0.96	0.85 (macro) / 0.93 (weighted)
Accuracy	-	-	93.10%
ROC-AUC	-	-	0.9637
Test loss (binary cross-entropy)	-	-	0.1793

The classifier achieved an overall accuracy of 93.10% and a ROC-AUC of 0.9637. The confusion matrix in Figure 3.8 shows that 48 of the 49 LSD-positive images were correctly identified, a sensitivity of 97.96%, while 6 of the 9 healthy images were correctly classified, a specificity of 66.67%. Only one infected animal was missing, while three healthy animals were incorrectly flagged as LSD-positive. The ROC curve in Figure 3.9 shows that the model reached a true positive rate of over 80% with no false positives at its most conservative operating points, and a true positive rate of 100% once the false positive rate reached about 33%.

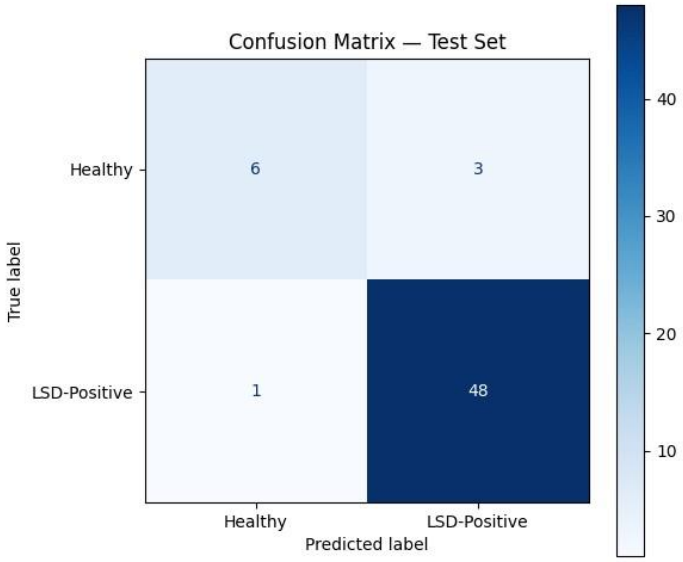


Figure 3.8: ResNet50 confusion matrix on the held-out test set

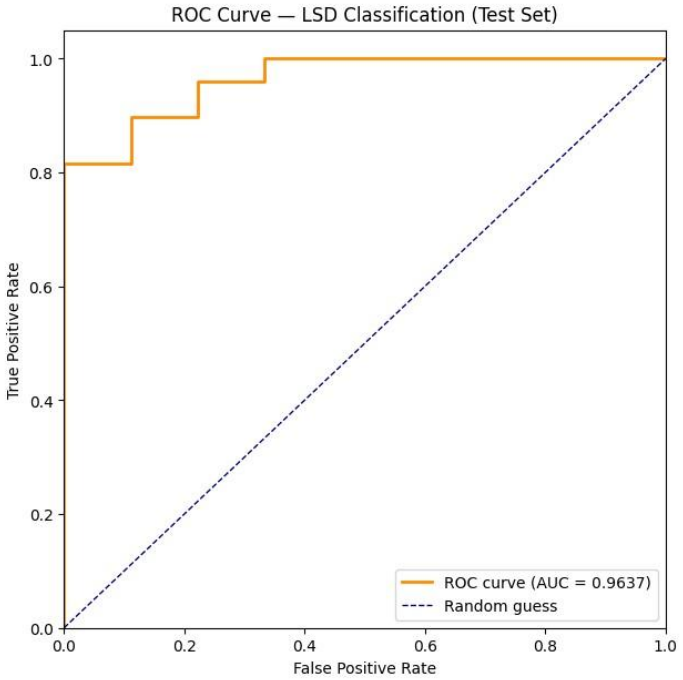


Figure 3.9: ResNet50 ROC curve on the held-out test set (AUC = 0.9637)

3.1.3 Integrated Detection-Classification Pipeline Results

The integrated pipeline combines the two models as described in Section 2.1.5. Two versions were evaluated on the 24 images of the detection test set, of which 18 were LSD-positive and 6 were healthy, using the decision threshold of 0.30. The first was the prototype pipeline developed in the training notebook (version 4), in which the overall probability was taken only from the highest region score, using a detection confidence threshold of 0.35. The second was the pipeline deployed in the CattleSense platform,

which adds the whole-image ResNet50 assessment, lowers the detection threshold to 0.10 and performs a second, more sensitive detection pass when needed. The deployed pipeline was evaluated by sending each test image to the running LSD microservice, which ran on a CPU-only Docker container on a development laptop. The results of both versions are compared in Table 3.4 and Figure 3.10.

Table 3.4: End-to-end performance of the integrated pipeline on the 24-image test set

Metric	Prototype pipeline (v4, notebook)	Deployed pipeline
Probability used for the decision	Highest regions score only	Maximum highest region scores and whole-image probability
Detection confidence threshold	0.35	0.10 (second pass at 0.06)
True positives / false negatives	4 / 14	18 / 0
True negatives / false positives	6 / 0	6 / 0
Accuracy	41.67%	100.00%
Recall (sensitivity) for LSD	22.22%	100.00%
Precision for LSD	100.00%	100.00%
F1-score for LSD	0.36	1.00
False positive rate	0.00%	0.00%
False negative rate	77.78%	0.00%
Average processing time per image	2.69 s (Tesla T4 GPU)	2.36 s (CPU)

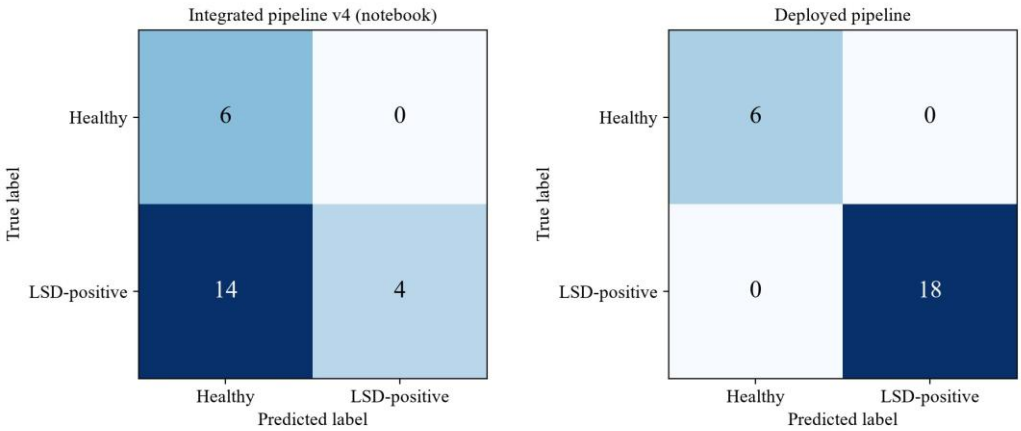


Figure 3.10: Prototype and deployed pipeline confusion matrices on the 24-image test set

The prototype pipeline was very cautious. It produced no false positives, but it missed 14 of the 18 infected animals, giving an accuracy of only 41.67% and a false negative rate of 77.78%. The reason is visible in the detector results above: whenever YOLOv8s found no nodule above the 0.35 threshold, the pipeline had no evidence and reported the animal as healthy.

The deployed pipeline classified all 24 test images correctly, identifying all 18 infected animals and all 6 healthy ones. The separation between the two classes was wide, as

shown in Figure 3.11. The lowest probability given to an LSD-positive image was 0.9486, and the median was 0.9999, so every infected animal was placed in the High-risk category. The highest probability given to a healthy image was 0.1481, so every healthy animal was placed in the Low-risk category, and no image fell into the Moderate band. The detector contributed regions for most infected animals, with a median of three detections per positive image, but on two of them (Lumpy_Skin_194 and Lumpy_Skin_197) it found no regions at all; these were identified correctly through the whole-image assessment. Only one healthy image produced a detection, with a low region score of 0.10, which did not affect its Low-risk classification. An important caveat on these figures is discussed in Section 3.3.3.

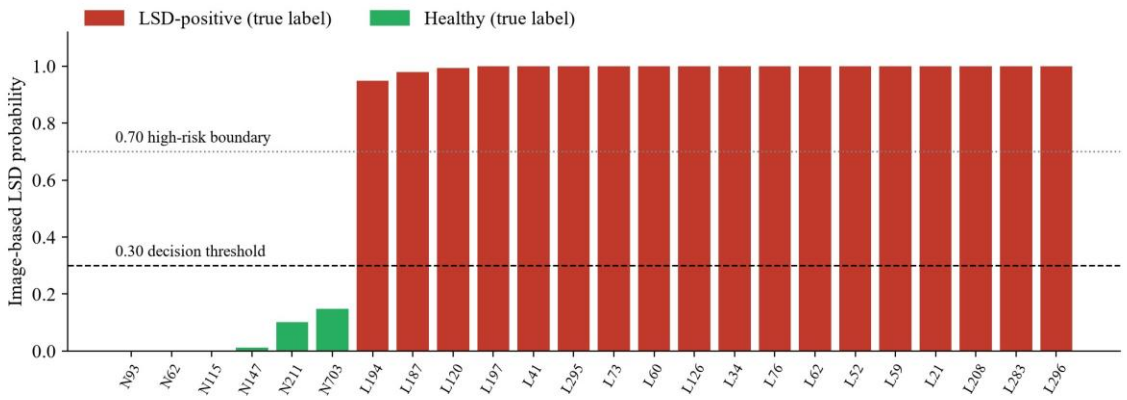


Figure 3.11: Image-based LSD probability for the 24 test images

Figure 3.12 shows the regions detected by the deployed pipeline on three test images. On the two infected animals the detector marked 22 and 15 regions respectively, most of them corresponding to visible nodules, and both animals were rated High risk. Healthy animals produced no detections and an overall probability of 0.27%.

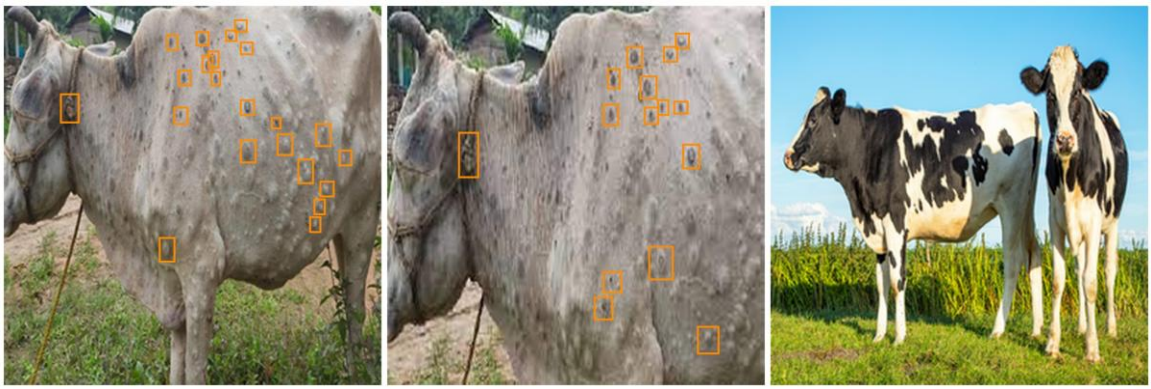


Figure 3.12: Detected regions in selected LSD test images.

The average processing time of the deployed pipeline was 2.36 seconds per image on a CPU, with a median of 1.17 seconds and a range of 0.50 to 11.67 seconds. As Figure 3.13 shows, the processing time depends almost entirely on the number of regions detected (correlation coefficient r = 0.996), because the ResNet50 classifier is run once for every detected region. This dependence becomes significant for images in which the low detection threshold produces many regions. During functional testing, a high-resolution image of 1536 by 1024 pixels produced 114 detected regions and required between 36 and 38 seconds to process, which exceeds the target of 10 to 15 seconds set in the proposal.

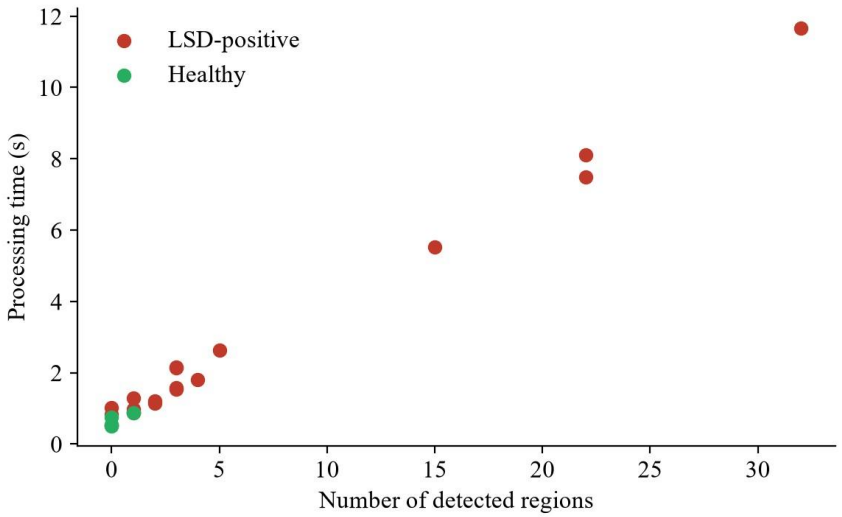


Figure 3.13: Relationship between the number of detected regions and processing time for the 24 test images

3.1.4 Hybrid Framework Results

The hybrid framework combines the image-based probability with the symptom-based probability using the 70:30 weighting described in Section 2.1.7. Since no dataset was available in which images were paired with recorded clinical signs, the fusion could not be evaluated for accuracy against confirmed diagnoses. Instead, its correctness and behavior were verified in two ways.

First, the functional tests reported in Table 2.8 confirmed that the deployed service computes the symptom probability and the fused probability exactly as specified. For example, an image with an image-based probability of 0.7864 combined with swollen lymph nodes, high fever and a body temperature of 40.5 °C produced a symptom probability of 0.5946 and a fused probability of 0.7289, which matches the formula 0.7 ×

0.7864 + 0.3 × 0.5946. The result card in Figure 2.10 shows the same calculation in the user interface, where a photo probability of 100% and a symptom probability of 73.0% combine to an overall probability of 91.9%.

Second, the effect of different symptom reports on the final decision was examined by submitting two test images to the deployed service with four different symptom profiles. The first image was the healthy test image with the highest image-based probability (Normal_Skin_703, 0.1481), and the second was the infected test image with the lowest image-based probability (Lumpy_Skin_194, 0.9486). The results returned by the service are listed in Table 3.5.

Table 3.5: Effect of different symptom reports on the fused result (outputs of the deployed service)

Test image	Symptoms reported	Image prob.	Symptom prob.	Final prob.	Risk level
Normal_Skin_703	No symptoms reported	0.1481	-	0.1481	Low risk
Normal_Skin_703	Non-specific signs only (reduced milk, decreased appetite)	0.1481	0.2258	0.1714	Low risk
Normal_Skin_703	Strong LSD signs (swollen lymph nodes, high fever, 41.0 °C)	0.1481	0.6216	0.2902	Low risk
Normal_Skin_703	Normal temperature only (38.5 °C)	0.1481	0.0000	0.1037	Low risk
Lumpy_Skin_194	No symptoms reported	0.9486	-	0.9486	High risk
Lumpy_Skin_194	Non-specific signs only (reduced milk, decreased appetite)	0.9486	0.2258	0.7317	High risk
Lumpy_Skin_194	Strong LSD signs (swollen lymph nodes, high fever, 41.0 °C)	0.9486	0.6216	0.8505	High risk
Lumpy_Skin_194	Normal temperature only (38.5 °C)	0.9486	0.0000	0.6640	Moderate risk

The results show that the fusion behaves as designed in straightforward cases. When no symptoms are reported, the image result is used unchanged. When strong LSD signs are reported for an infected animal, the result remains in the High-risk category. The table also reveals two less obvious properties of the design. Reporting strong LSD signs for the healthy-looking image raised its probability from 0.1481 to only 0.2902, which is still below the 0.30 threshold and therefore remains Low risk. Conversely, reporting only a normal body temperature for the infected animal lowered its probability from 0.9486 to 0.6640, moving it from High to Moderate risk, because the unticked symptoms were counted as absent. Both properties are examined in Section 3.3.4.

3.2 Research Findings

3.2.1 Findings of YOLOv8s Detection

The YOLOv8s detector learned to recognize LSD nodules to some extent, but its performance as a stand-alone detector was low, with a test mAP@0.5 of 11.83%, precision of 19.05% and recall of 27.59%. The training curves showed clear overfitting after about 25 epochs, with the validation box and distribution focal losses rising while the training losses continued to fall. At the level of individual nodules, the detector missed most of the annotated lesions in the validation set. The detector was nevertheless able to localize visible nodules on many infected animals, and it produced very few detections on healthy animals: only one of the six healthy test images produced any detection in the deployed pipeline. Its value in the final system lies mainly in localizing lesions for the user and in supplying candidate regions to the classifier, rather than in deciding on its own whether an animal is infected.

3.2.2 Findings of ResNet50 Image Classification

The ResNet50 classifier was the strongest single component of the system. On a clean, held-out test set it reached an accuracy of 93.10% and a ROC-AUC of 0.9637, with a sensitivity of 97.96% for LSD-positive images. Its main weakness was specificity: three of the nine healthy test images were incorrectly classified as LSD-positive, giving a specificity of 66.67%. Two-phase transfer learning was effective, with the model converging within a few epochs, although the difference between training and validation loss indicates some overfitting. The small number of healthy images, only nine in the test set, means that the specificity estimate is uncertain.

3.2.3 Findings of the Integrated Detection-Classification Pipeline

The way in which the two models were combined had a decisive effect on the end-to-end result. The prototype pipeline, which depended entirely on the detector, missed 77.78% of infected animals. Adding the whole-image ResNet50 assessment and using a more sensitive detection threshold in the deployed pipeline eliminated these missed cases on the test set, with all 24 images classified correctly and a wide margin between the two classes. The deployed pipeline processed images in 2.36 seconds on average on a CPU, but its processing time grows with the number of detected regions, and it can exceed the proposal's target for images that produce many regions.

3.2.4 Findings of Hybrid Multi-Modal Fusion

The hybrid fusion was implemented correctly and produced exactly the expected values in every test. It makes the reasoning behind each result transparent, since the user can see how much the photograph and the reported symptoms each contributed. However, the analysis of different symptom profiles showed that, with the current 70:30 weighting, the symptom checklist has only a limited influence on the final decision. On its own it cannot raise a healthy-looking image into the Moderate risk category, and a partially completed checklist can lower the result for an animal whose photograph clearly shows LSD.

3.2.5 Overall System Findings

Taken together, the results show that the system's ability to identify infected animals comes mainly from the ResNet50 classifier, supported by the detector's localization of lesions, while the symptom branch adds transparency and clinical context rather than a large change in the decision. The complete component was successfully deployed as a working microservice within CattleSense, with a web interface, risk-based guidance, annotated images and PDF reports, and it handled invalid input and partial failures correctly. The main areas requiring further work are the accuracy of the nodule detector, the independence of the evaluation data, the processing time for images with many detections, and the calibration of the symptom fusion.

3.3 Discussion

3.3.1 Discussion of Detection Performance

The low detection scores need to be understood in the context of the data. LSD nodules are among the hardest targets for an object detector. They are small, with a median box covering only about 6% of the image width; they are numerous, often with dozens in a single image; they frequently touch or merge with one another; and their colour and texture vary with the coat of the animal. Dense collections of small, similar-looking objects are a well-known weakness of single-stage detectors, and evaluation with mAP is particularly strict in such cases, because a box that is only slightly misplaced on a small object fails the 50% overlap requirement.

A second and probably more important factor is the completeness of the annotations. As Figure 3.2 shows, in several training images only some of the visible nodules were labelled, and one composite image contained a second animal whose nodules were not

labelled at all. Incomplete annotation harms a detector in two ways. During training, the model is penalized for correctly detecting a nodule that happens not to be labelled, which teaches it to be less confident about genuine lesions. During evaluation, such correct detections are counted as false positives, while every labelled nodule it misses is counted as a false negative. Part of the 120 false positives in Figure 3.5 is therefore likely to consist of real but unlabeled nodules, and the true precision of the detector is probably higher than the measured value. Annotating a dense lesion pattern consistently is genuinely difficult, and the annotation work was divided between two people, which may have increased inconsistency.

Two further factors contributed. The detection dataset was small, with only 182 original training images, which explain the early overfitting seen in Figure 3.3. In addition, the training images were processed in Roboflow with adaptive histogram equalization, which, as Figure 3.2 shows, introduced strong changes in colour and texture, whereas the deployed service does not apply the same step to uploaded photographs. A mismatch between training and inference preprocessing means that the detector sees images in use that look different from those it was trained on. The original images in the dataset are also low in resolution, typically 256 by 256 pixels, which leaves very few pixels for each nodule once the image is enlarged to 640 by 640 pixels.

Despite these limitations, the detector still provides value in the final system. It localizes lesions so that the user can see what the system found, it supplies candidate regions to the classifier, and it rarely produces detections on healthy animals. Improving it is mainly a matter of data rather than architecture: complete and consistent annotation, more original images, higher-resolution photographs and consistent preprocessing between training and deployment are likely to bring much larger gains than changes to the model itself.

3.3.2 Discussion of Image Classification Performance

The ResNet50 classifier performed well, and its accuracy of 93.10% is within the range reported in the literature for image-level LSD classification. Classifying a whole image is a considerably easier task than localizing every nodule: the classifier only needs to recognize the overall appearance of affected skin, which is distinctive in most LSD cases, and it benefits directly from features learned on ImageNet. This explains why it converged within a few epochs and why it is the more reliable of the two models.

The classifier's errors were not evenly distributed. It was highly sensitive, missing only one infected animal, but less specific, misclassifying three of nine healthy animals. This pattern is consistent with the class imbalance in the dataset, which contains about five LSD images for every healthy one, so the model sees far more examples of affected skin than of healthy skin. Some healthy cattle also have naturally mottled or patchy coats that can resemble nodular skin at low resolution. In addition, at least one image in the healthy class (Normal_Skin_5) appears to show nodules, which suggests that the dataset contains some label noise. For a screening tool, a bias towards sensitivity is acceptable, since a false alarm leads only to closer inspection, whereas a missed case can allow the disease to spread. However, the specificity estimate is based on only nine healthy test images, so it should be confirmed with a larger and more balanced test set.

3.3.3 Discussion of the Integrated Pipeline Performance

The improvement from 41.67% to 100% accuracy between the prototype and deployed pipelines is the most striking result of this chapter, and it reflects a deliberate design change. The prototype relied on the detector to provide all the evidence, so the detector's low recall translated directly into missed cases. The deployed pipeline treats the detector and the classifier as complementary: the classifier's assessment of the whole image acts as a safety net when the detector misses small or diffuse lesions, while the detector continues to localize lesions whenever it can. This combination proved robust on the test images, and the absence of any image in the Moderate band shows that the probabilities were well separated.

The 100% result must, however, be interpreted with caution. The YOLOv8s test set and the ResNet50 dataset split were created independently, from the same pool of images. The 24 images used to evaluate the pipeline were never seen by the detector during training, but many of them are likely to have been part of the ResNet50 training set, which contained 70% of all images. Since the deployed pipeline relies heavily on the classifier's whole-image probability, some of its performance on these images may come from the classifier having seen them before. The figure of 100% should therefore be regarded as an optimistic upper bound. The most reliable estimate of the system's ability to recognize unseen LSD images remains the ResNet50 result on its own clean test set, an accuracy of 93.10% and a sensitivity of 97.96%. A definitive evaluation of the full

pipeline would require a common split in which the same test images are excluded from the training of both models, and a new set of independently collected images.

The processing time results also deserve comment. The average of 2.36 seconds on a CPU shows that the system is fast enough for interactive use with typical images. However, because the classifier is run on every detected region, the low detection threshold that improves sensitivity can also produce a very large number of regions on busy or high-resolution images, which raised the processing time to more than 30 seconds in the most extreme case observed. Limiting the classifier to the highest-scoring regions, processing regions in batches, or resizing large images before detection would keep the processing time within the target without reducing the system's sensitivity.

3.3.4 Discussion of Hybrid Fusion Effectiveness

The hybrid fusion achieves its main purpose of combining two independent sources of evidence in a transparent way. The decision to exclude nodule-related questions from the checklist ensures that the image and the symptoms do not double-count the same evidence, and the breakdown displayed to the user makes the basis of each result clear. Verifying the fusion through controlled tests showed that it is implemented exactly as specified.

The analysis in Table 3.5 shows, however, that the current weight limits the practical influence of the symptoms. Because the symptom branch contributes at most 30% of the final probability, a full set of classic LSD signs reported for an animal whose photograph looks healthy can raise the result to just below 0.30 but not beyond it. This is significant because fever and swollen lymph nodes can appear before nodules become visible, which is exactly the situation in which symptom information would be most valuable. In addition, scoring treats every unticked symptom as absent once any symptom has been reported, so a farmer who records only a normal temperature for an obviously infected animal lowers its risk level. A more effective design would distinguish between symptoms that are absent and symptoms that were simply not assessed and would allow strong clinical signs to raise the risk level on their own, for example by flagging any animal with both high fever and swollen lymph nodes for veterinary attention regardless of the image. The weights themselves are based on published clinical descriptions of LSD rather than on data and calibrating them would require a dataset of images paired with recorded clinical signs and confirmed diagnoses.

3.3.5 Comparison with Existing Studies

Table 3.6 compares the results of this research with the LSD detection studies reviewed in Chapter 1. Direct comparison of accuracy figures should be made with care, since each study used a different dataset, a different number of images and a different evaluation protocol.

Table 3.6: Comparison of the proposed system with existing LSD detection studies

Study	Method	Dataset size	Reported result	Localization	Symptom fusion	Deployed tool
Genemo (2023) [10]	CNN features + ELM	Not stated in summary	~90% accuracy	Partial	No	No
Saqib et al. (2024) [7]	MobileNetV2 + RMSProp	793 images	95% accuracy	No	No	No
Saha (2024) [9]	MobileNetV2 and others	840 images	96% accuracy, AUC 0.98	Partial	No	No
Senthilkumar et al. (2024) [8]	Ten pretrained CNNs	3,247 images	~96% accuracy	No	No	No
Ayon et al. (2026) [11]	Ensemble of three CNNs (LSD and FMD)	10,516 images	98.2% accuracy	No	No	No
This study	**YOLOv8s + ResNet50 + symptom fusion**	**382 (classification), 254 (detection)**	**ResNet50: 93.10% accuracy, AUC 0.9637; YOLOv8s: mAP@0.5 11.83%**	**Yes**	**Yes**	**Yes**

The classification accuracy of this study is slightly lower than the figures of 95% to 98% reported in the other studies. This is not surprising given that its classification dataset is the smallest of those compared, with only 62 healthy images, and that larger and more balanced datasets generally produce more accurate and better calibrated classifiers. The most recent study, by Ayon et al., used more than ten thousand expert-annotated images from three countries, roughly twenty-five times more data than was available here.

Where this research differs from the existing studies is in scope rather than in raw accuracy. None of the studies compared localized individual nodules with a trained object detector, none combined image evidence with clinical symptoms, and none produced a working tool that farmers can use. This study delivers all three, together with risk-based guidance and a report that can be shared with a veterinarian. The detector's results also highlight a challenge that classification-only studies do not face, namely that localizing

every nodule is substantially harder than recognizing an infected animal, and that progress in localization depends heavily on the quality and completeness of the annotations.

3.3.6 Limitations of the Proposed System

Limited and imbalanced data. The classification dataset contained only 382 images, of which just 62 were healthy, and the detection dataset contained 254 original images. The test sets were correspondingly small, with only nine healthy images in the classification test set, which makes the reported metrics sensitive to individual images and limits how far they can be generalized.

Incomplete and inconsistent annotation. Many visible nodules were not annotated, and some composite images contained unlabeled animals. This lowered both the training quality and the measured performance of the detector.

Low detection accuracy. The YOLOv8s detector achieved a test mAP@0.5 of 11.83%, far below the 80% target, and showed clear overfitting. It cannot yet be relied upon to find every nodule in an image.

Overlapping evaluation data. The detection and classification splits were created independently, so the end-to-end evaluation of the deployed pipeline may include images that the classifier saw during training. Its 100% accuracy is therefore an optimistic estimate.

Preprocessing mismatch. Adaptive contrast enhancement was applied to the detection training images but is not applied to images analyzed by the deployed service.

Data source and image quality. The images are mostly low-resolution photographs of 256 by 256 pixels gathered from various sources, and they were not collected specifically from Sri Lankan farms. Performance on photographs taken by local farmers with their own phones may differ. At least one image also is mislabeled.

Uncalibrated symptom fusion. The symptom weights are based on clinical literature rather than on data, the 70:30 weighting prevents strong symptoms from raising the risk level on their own, and unticked symptoms are treated as absent rather than unknown.

Processing time on busy images. Processing time increases with the number of detected regions and exceeds the target for an image with 114 detected regions.

No clinical or differential validation. The system has not yet been tested against veterinary diagnoses in the field, and it has not been evaluated on other skin conditions such as dermatophilosis, ringworm or insect bites, which could be confused with LSD.

Despite these limitations, the research demonstrates that a hybrid system combining nodule detection, image classification and symptom assessment can be built, deployed and used in practice to support early LSD screening. The limitations identified here point clearly to the next steps: better and more completely annotated data, a common evaluation split for both models, field validation with veterinary officers, and refinement of the symptom fusion.

4. CONCLUSION

This research set out to develop the Lumpy Skin Disease detection component of CattleSense, a multimodal platform for the early detection of bovine diseases. The motivation was a practical one. LSD spreads quickly among cattle, yet in Sri Lanka its diagnosis still depends on a veterinarian examining the animal in person, which is often not possible early enough in rural areas. The aim was therefore to give farmers a tool that could assess a photograph of an animal, consider the clinical signs they observe, and tell them clearly how likely it is that the animal is infected and what they should do next.

To achieve this, a hybrid system was designed and built around two deep learning models and a clinical scoring module. A YOLOv8s detector was trained to locate individual skin nodules, and a ResNet50 classifier was fine-tuned through transfer learning to estimate the probability of LSD for each detected region and for the image. These were combined with an optional symptom checklist covering signs that a camera cannot capture, such as swollen lymph nodes and fever, through a weighted fusion in which the image contributes 70% and the symptoms 30%. The final probability is translated into a Low, Moderate or High-risk level with practical guidance, an annotated image of the detected nodules and a downloadable PDF report in English or Sinhala. The complete component was deployed as a containerized Flask microservice within the CattleSense web platform, with a bilingual interface, secure access and a record of assessments for each animal.

The evaluation gave a mixed but informative picture. The ResNet50 classifier was the strongest component, reaching an accuracy of 93.10%, a ROC-AUC of 0.9637 and a sensitivity of 97.96% on its head-out test set. The YOLOv8s detector performed poorly as a stand-alone model, with a mAP@0.5 of 11.83%, and the analysis traced this to the small size of the detection dataset, the very small and densely packed nodules, and incomplete annotation of the training images. The way the two models were combined proved decisive. A prototype pipeline that relied on the detector alone missed 77.78% of infected animals, whereas the deployed pipeline, which adds a whole-image assessment as a safety net, classified all 24 test images correctly with an average processing time of 2.36 seconds on a CPU. Because some of these test images may have been seen by the classifier during training, this result is treated as optimistic, and the classifier's 93.10% on unseen images remain the more reliable estimate. The symptom fusion worked exactly as

specified and made each result transparent to the user, although its current weighting gives the clinical signs only a limited influence on the final decision.

Measured against the objectives set out in Chapter 1, the research prepared annotated datasets for detection and classification, developed and evaluated both deep learning models, integrated them into a single pipeline, incorporated clinical symptoms through hybrid fusion and risk stratification, and delivered the result through a working web platform. The main objective that was not fully met was the detection accuracy target, and the reasons for this have been identified clearly. Compared with existing LSD studies, which classify whole images and stop reporting experimental results, this work contributes nodule-level localization, the fusion of visual and clinical evidence, risk-based decision support and deployed tools that farmers can actually use.

Several directions for future work follow directly from the findings of this study:

- **Improving the detection data.** Re-annotating the detection dataset so that every visible nodule is labelled consistently, removing composite images, adding more original and higher-resolution photographs, and applying the same preprocessing during training and deployment.
- **Collecting local data.** Gathering images of LSD-affected and healthy cattle from Sri Lankan farms and veterinary offices, with confirmed diagnoses, to train and test the models on the conditions in which they will be used.
- **A common evaluation split.** Evaluating both models and the full pipeline on the same set of images that neither model has seen, to obtain a definitive measure of end-to-end performance.
- **Refining the symptom fusion.** Distinguishing between symptoms that are absent and symptoms that were not assessed, allowing strong clinical signs to raise the risk level on their own, and calibrating the weights using images paired with recorded clinical signs.
- **Differential diagnosis.** Extending the system to distinguish LSD from conditions with similar appearance, such as dermatophilosis, ringworm and insect bites.
- **Efficiency and field validation.** Limiting classification to the most promising regions to keep processing time low on busy images and validating the system prospectively with veterinary officers in the field.

In conclusion, this research has shown that a hybrid approach combining nodule detection, image classification and clinical symptom assessment can be built and deployed as a practical screening tool for Lumpy Skin Disease. While the detector requires further work, the system already offers farmers an accessible and transparent way to assess their animals and to seek veterinary help sooner. With better data and the refinements identified here, it has the potential to make a real contribution to the early control of LSD and to the livelihoods of the farmers who depend on their cattle.

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